Analog IC Design Implementation and Verification

Introduction to Analog IC Design and Processes – Week 1 Labs

Course Version IC 23.1 ISR11.29 with SPECTRE 24.1 ISR1

Lab Manual **Revision 1.0**

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Introduction to Analog IC Design and Processes

Module 1: Design and Characterization of an n-MOS Circuit

Lab 1-1 Design of an n-MOS Circuit

Objective: To use the basic schematic capture commands to build an n-MOS circuit.

In this lab, you open the schematic capture environment window and add components, wires and pins. The schematic is then checked and saved.

Creating a Library and Cellview

1. Open the Terminal window, open the directory of **basic_analog_labs,** by entering this command in the terminal window:

   ```
   cd basic_analog_labs
   ```

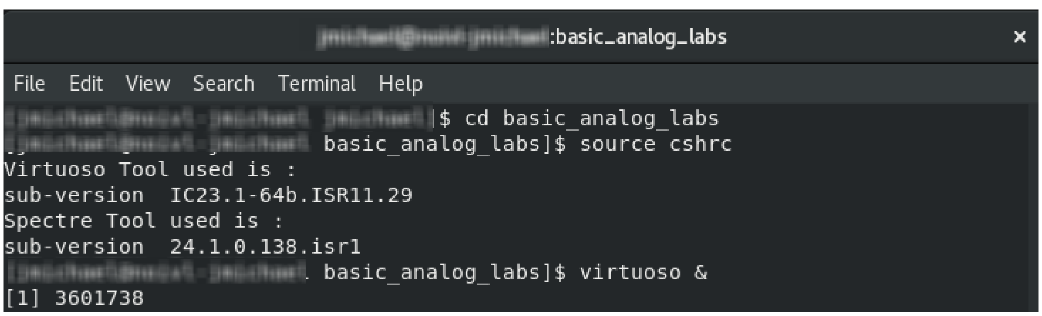
2. In that directory, source the License file by entering this command in the terminal window:

   ```
   source cshrc
   ```

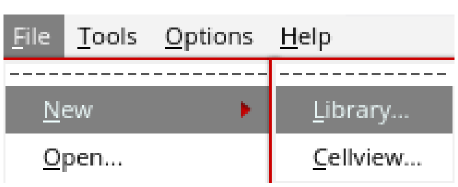
3. Type the following command to open CIW:

   ```
   virtuoso &
   ```

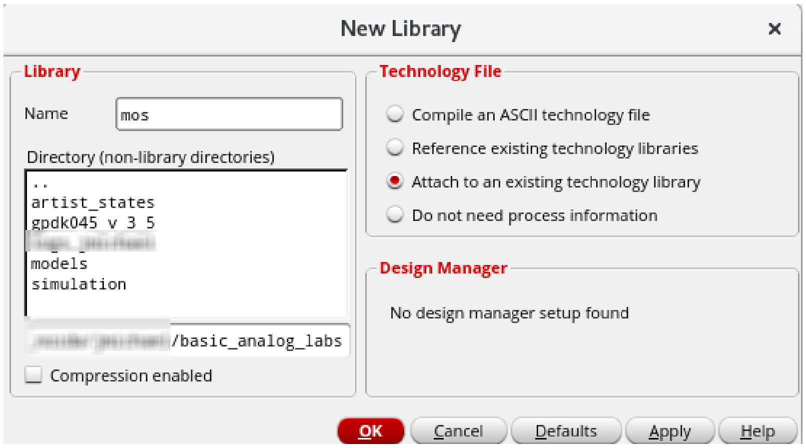
   The following screenshot gives the view of the terminal window, on these commands executed:



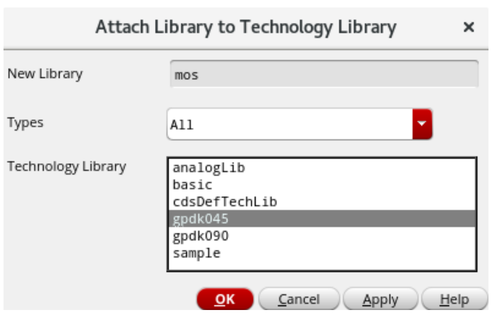
4. Open **CIW** and from CIW, choose **File – New – Library.**



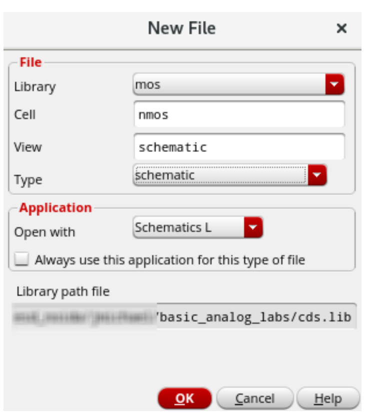
5. Name the Library as **mos** and select the radio button for **Attach to an existing technology Library** and click on **OK.**



6. In the Attach Library to Technology Library form, select **gpdk045** library and click **OK**.



7. Create a new cell view under the same **mos** library by choosing **File – New – Cellview** and name it as **nmos**, then click **OK** to open the schematic view.



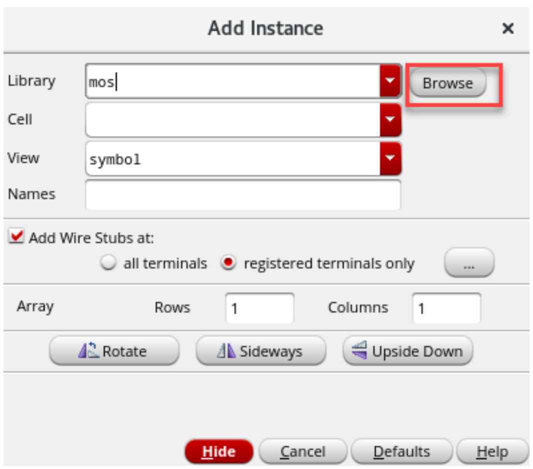
Invoking the Add Instance Form in the Cellview

The instructions in this section describe how to build the n-mos circuit schematic shown in the following snapshot.

1. With your cursor in the schematic window, go to the top of the schematic window to the toolbar. Click the **Create Instance** icon to display the Add Instance form. 

Note: You can also choose **Create – Instance** from the pull-down menus or use the **I** key to pop up the form.

2. In the **Add Instance** form, click on **Browse** to open the Library Browser.



3. In the **Library Browser – Add Instance** form, select **gpdk045** in the library column. The Cell column of this form now displays the available cells in that library.

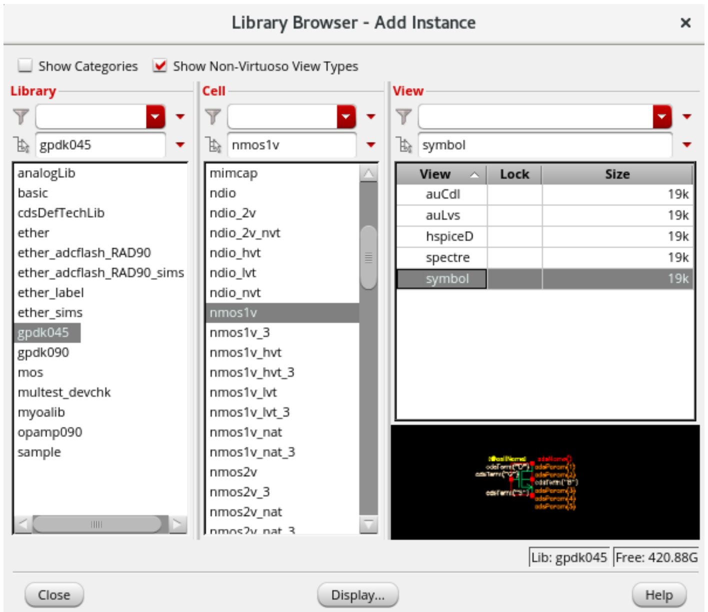
   a. Use the scroll bar to the right of the Cell column to locate **nmos1v** and **click** to select this cell. The Library Browser form now displays several views for the *nmos1v* cell.

   Note: You can use the filtering option under the Cell category.

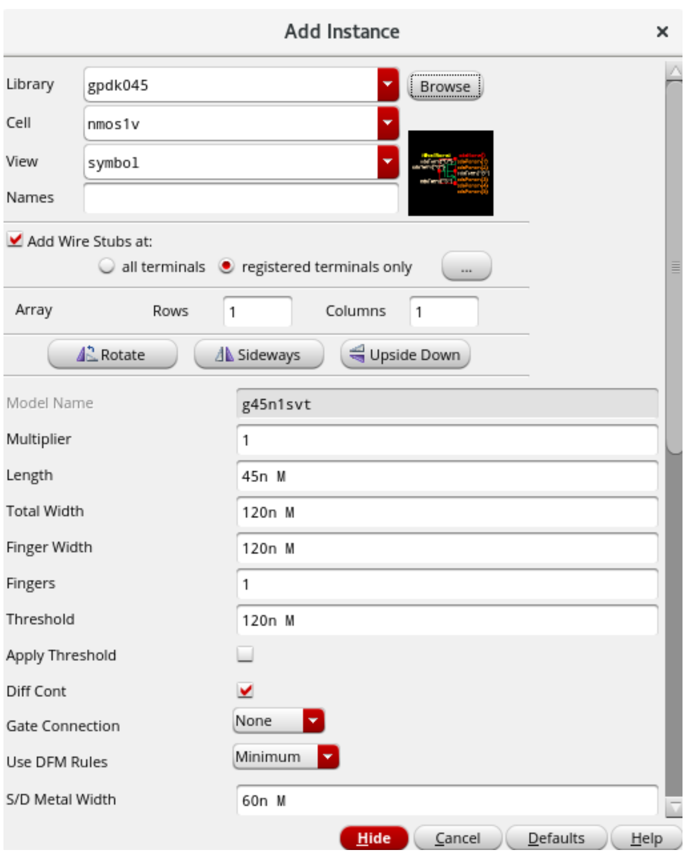
   Note: You can also begin typing ***nmos1v*** in the *cell* name field of the Add Instance form. The auto-complete feature brings the *nmos1v* cell name into the field for easy selection.

   As you make your selection, note that the Add Instance form modifies itself.

b. Select the **symbol view** with the left mouse button. Your Library Manager should be like what you see in the following snapshot.



The Add Instance form now displays some information along with auto-filled text fields, as shown in the following snapshot.



4. Close the **Library Browser – Add Instance** form. Leave the Add Instance form open with *Library*, *Cell* and *View* pull-down cyclic fields populated.

Adding Components to a Schematic

1. Click the **Hide** button toward the bottom of the form.

   Follow the next steps diligently. They will allow you to place the chosen *nmos* transistor on your schematic.

   a. Move your cursor into the schematic window. A highlighted symbol is displayed. It is ready to be placed on the schematic.

   b. **Left-**click to place the symbol on the schematic.

   c. Place the instance of *nmos1v* on the schematic.

      Note: Observe that the device has been added to the schematic and the symbol has an instance name. A highlighted symbol of *nmos1v* remains and moves with the cursor.

   d. Click the **<Esc>** key to quit placement mode.

2. Press the **I** key.

   The Add Instance form is displayed again, with the library, cell, and view fields filled. Likewise, a highlighted symbol tracks the cursor in the schematic window.

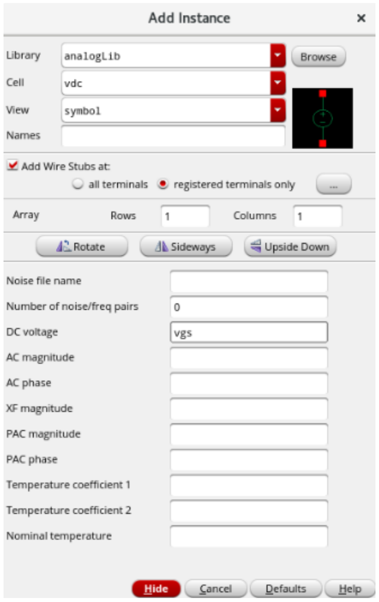
   a. Move the cursor into the Add Instance form. Change the library field to **analogLib** from the library column. Click in the cell text field and choose **vdc**.

      Note: You can use the **Browse** button to search for *vdc*. However, entering *vdc* directly into the text field saves time.

   b. Set the parameter as follows:

      DC voltage: `vgs`

c. Your form should be similar to what you see in the following snapshot:



d. **Hide** the form. Place this instance of *vdc* on the schematic, on left side of *nmos1v* instance.

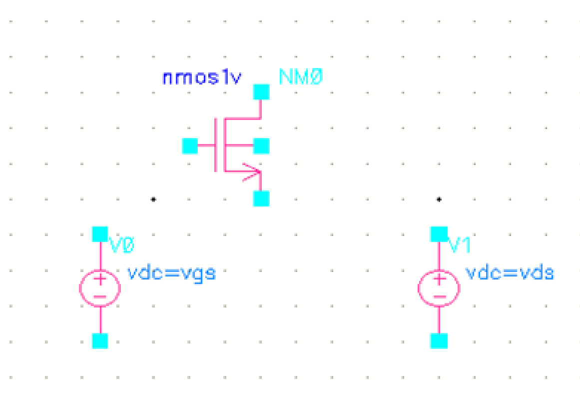
3. Press the **I** key.

   With the already selected **vdc** in the Add Instance form, change the parameter as follows:

   DC voltage: `vds`

   Hide the form. Place this instance of *vdc* on the schematic, on right side of *nmos1v* instance.

4. You can use the keyboard arrow keys to move up and down the schematic. Your canvas should be similar to what you see in the following snapshot.



Adding Ground to the Schematic

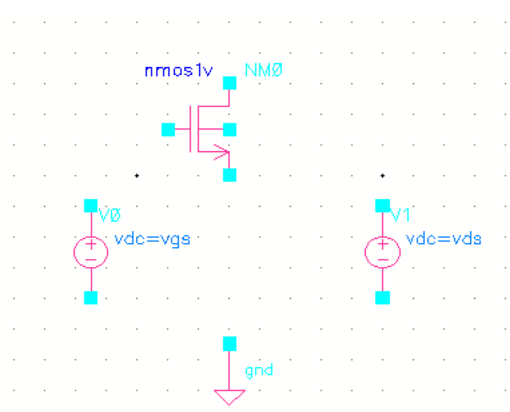
1. Zoom out with the **[** key, so you have room to place the power and ground symbols.

2. Press **I** to display the Add Instance form.

3. In the Cell text field of the Add Instance form, replace *vdc* with **gnd** under the same *analogLib* library.

4. Move the cursor into the schematic and place the *gnd* symbol below the source of the *nmos1v* device. Then press the **Esc** key. This will *close* the Add Instance form.

   Important: Pressing **Esc** cancels any active command in the schematic window.



Adding Wires to a Schematic

Add wires to connect components in the design.

1. Choose **Create – Wire (narrow)** from the pull-down menu. You can also use the **Create Narrow Wire** button from the Create toolbar section.

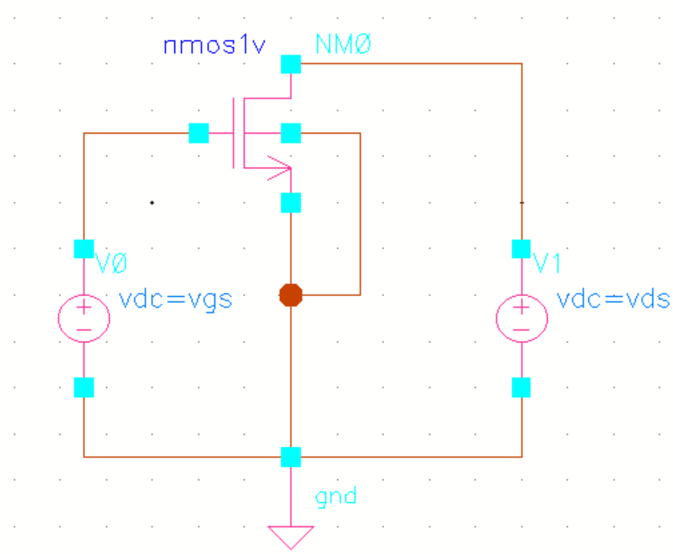
   Note: Lowercase **w** is the bindkey for a *Narrow wire*, **Shift-W** for a *Wide wire*.

2. In the schematic window, click one of your components' terminals, such as the *vdc* positive terminal, to make it the first point for your wiring.

   Note: The wire router cannot always find a path between two points. If this happens, a dotted flight line is drawn between the points and establishes connectivity. If flight lines occur in your lab, delete them and click on the intermediate points to guide the router as your design is wired.

   Follow the prompts at the bottom of the design window, move the mouse cursor and **left-**click the destination point for your wire, such as nmos gate. A net is routed.

3. Continue wiring the schematic, as shown in the following snapshot.



   The exact topology shown in the reference schematic is not as important as the connectivity.

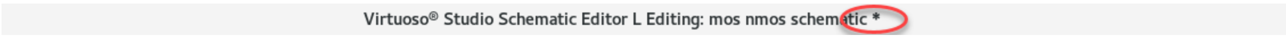
   Caution: *Be sure to connect the substrate for the nmos1v transistor to gnd respectively.*

   Important: If you make a mistake while wiring, select that section of the wire and press the **<Delete>** key to delete that part of the wire.

4. Press **Esc** with your cursor in the schematic window to cancel the wiring when finished wiring.

Saving a Design

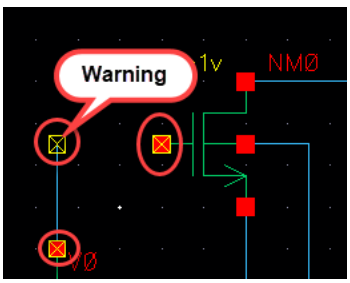
Note on the title bar that there is an asterisk next to the name schematic. This indicates that your schematic has been edited and needs to be saved.



1. Click the **Check and Save** icon in the schematic editor window.

2. Or choose **File – Check and Save** or click **Shift + X**.

Note: A yellow box appears around all the unconnected pins and wires. If there are any, close the associated warning message and complete the connectivity.

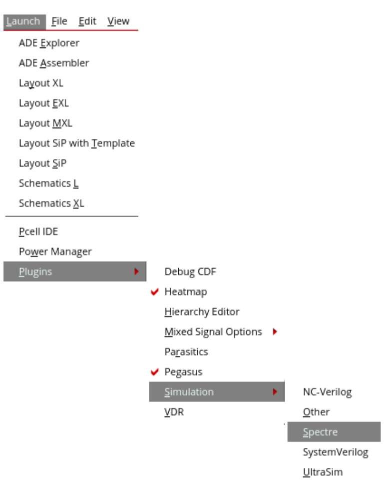


3. Check the CIW output area for any messages after a successful check and save.

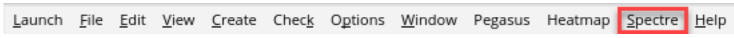
```
INFO (SCH-1170): Extracting "nmos schematic"
INFO (SCH-1426): Schematic check completed with no errors.
INFO (SCH-1181): "mos nmos schematic" saved.
```

Netlisting the Design

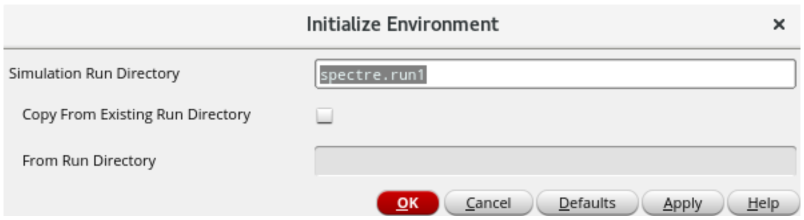
1. From the Schematic Editor window, choose **Launch – Plugins – Simulation – Spectre**.



Note: This attaches the Spectre Classic Simulator in the menu bar.

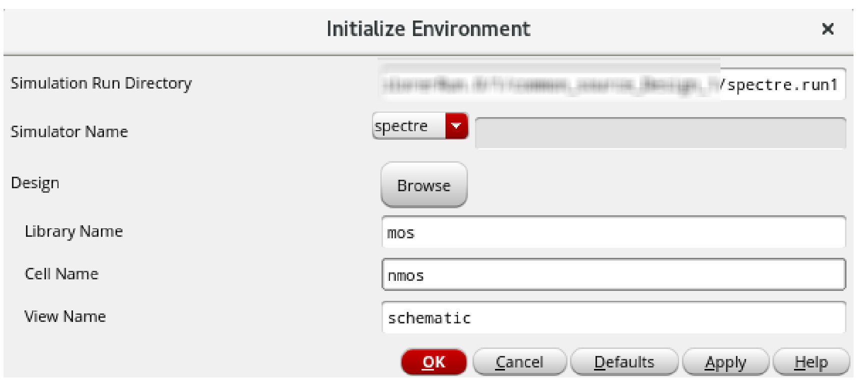


2. From the menu bar, choose **Spectre – Initialize**.

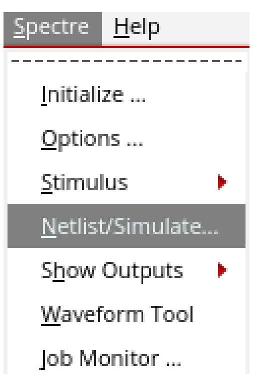


3. Click **OK** in the Initialize Environment form to create simulation result databases in the *spectre.run1* default simulation run directory.

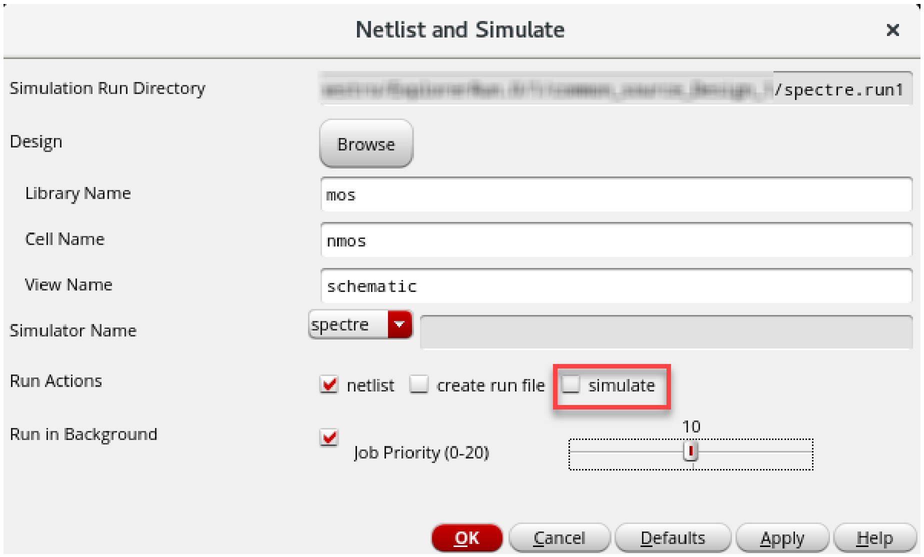
   a. Click **OK** in the next form that opens (shown in the following snapshot), verifying the design to be netlisted.



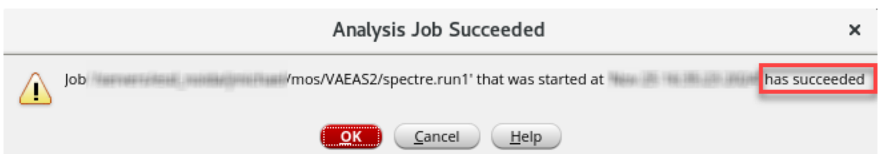
4. Choose **Spectre – Netlist/Simulate** to open the Netlist and Simulate form.



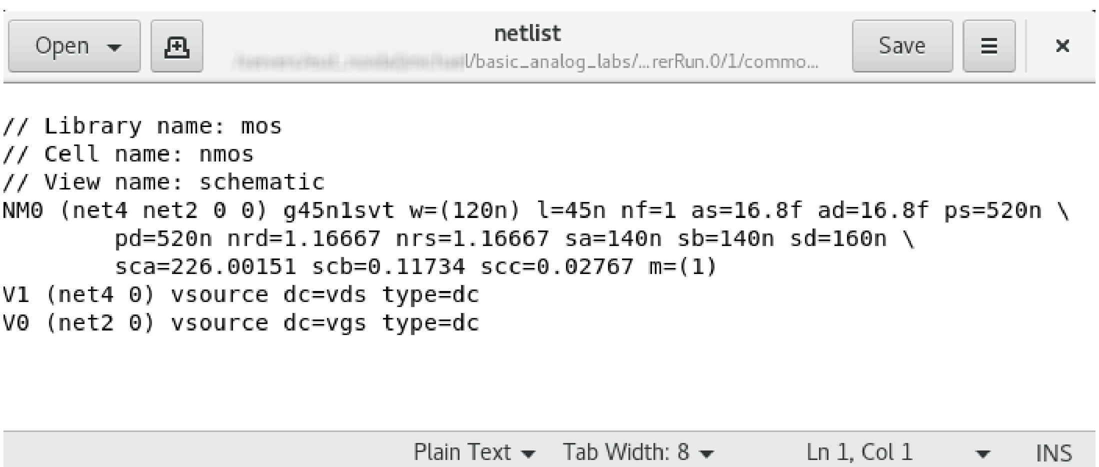
5. **Disable** the *Run Actions – simulate* option. Your form should look similar to the one you see in the following snapshot.



6. Click **OK**. In the *Analysis Job Succeeded* form that comes up, click **OK**.



7. View the netlist file in the *spectre.run1* directory by choosing **Spectre – Stimulus – Edit Netlist File**.



Try to associate the connectivity in the netlist along with your drawn schematic.

8. **Close** the netlist window, leave *nmos* schematic view and CIW session running, and proceed to the next lab.

End of Lab

Lab 1-2 Simulation Using ADE Explorer and Virtuoso VA

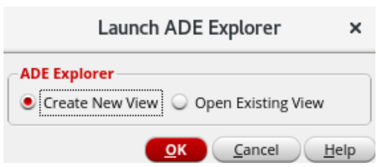
Objective: To simulate the n-MOS circuit using ADE Explorer and Virtuoso VA.

In this lab, you

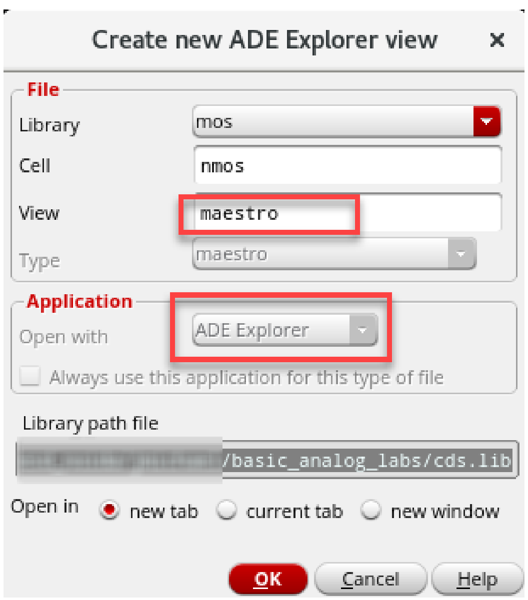
- Learn how to simulate the n-MOS created in the previous lab by creating a new *maestro view* using Virtuoso® ADE Explorer.

Starting the Virtuoso ADE Explorer and Creating a *maestro* View

1. Choose **Launch – ADE Explorer** from the *nmos* schematic window to open the Launch ADE Explorer form.

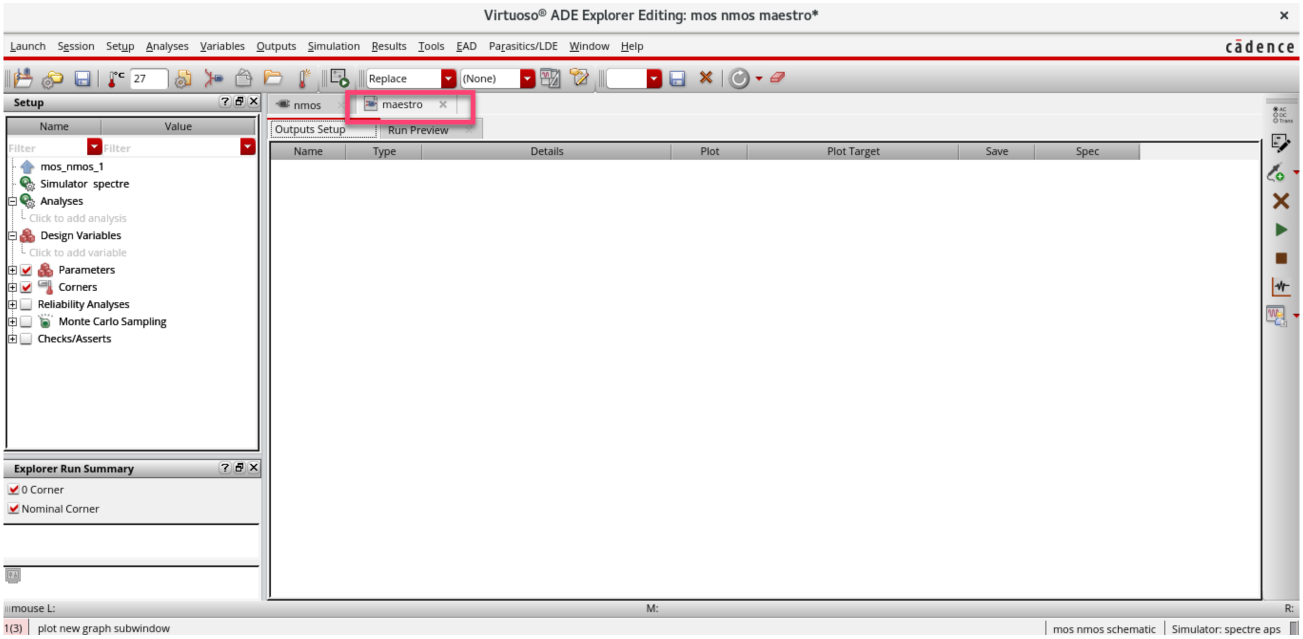


2. Select **Create New View** to open the **Create new ADE Explorer view** form.



3. The **maestro** view type is chosen by default, and so is the *Open with* the application, which is set to the **ADE Explorer**.

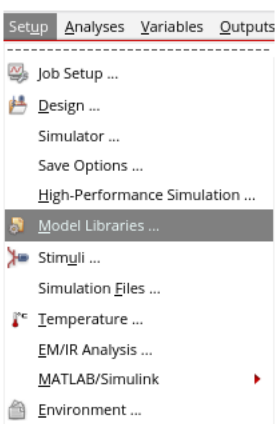
   Leave the View name set to the default name, **maestro**. Choose to open in a new tab and click **OK** on this form to open the ADE Explorer environment in a new **maestro** tab.



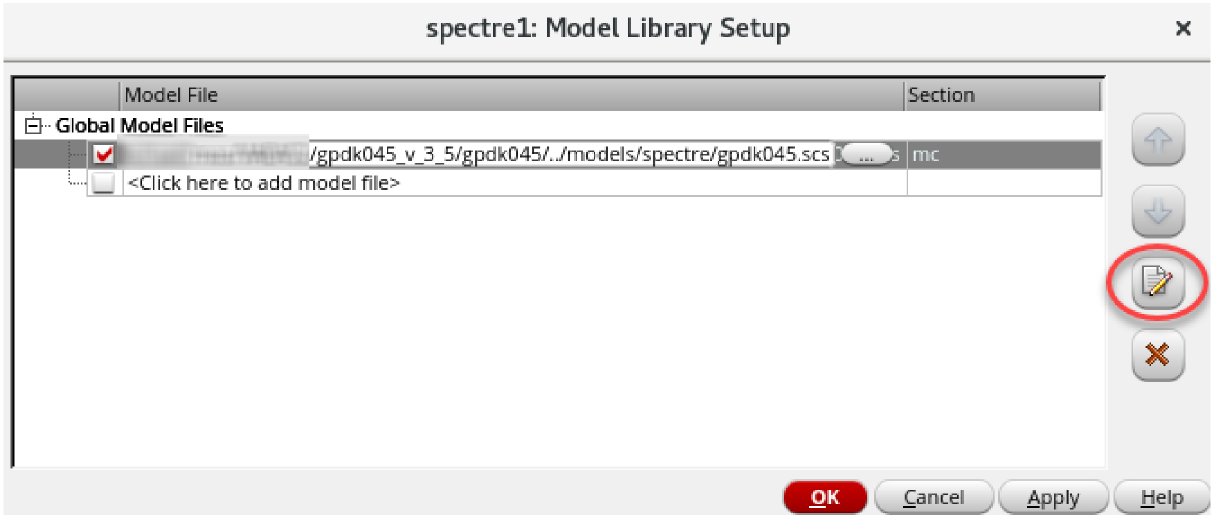
Setting the Model Libraries

1. The model library file contains the model files that describe the *nmos1v* device during simulation.

   a. Choose **Setup – Model Libraries**.



The Model Library Setup form appears. You can see that there will be *gpdk045 model library* added automatically. This is because, we have attached the **mos** library to *gpdk045* in the beginning.



b. To view the model file, select the model file. Click the **Edit/View Selected Files** button.

Close the model file after viewing.

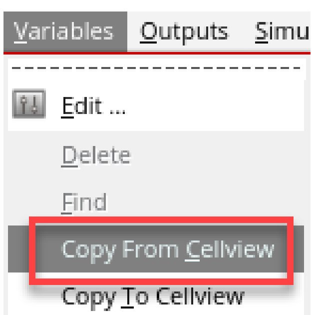
With the Model Library Setup form, you can include multiple models.

c. On the Model Library Setup form, click **OK.**

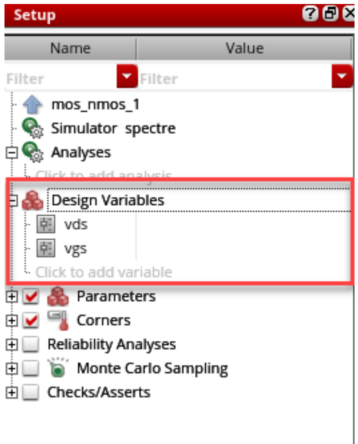
Setting Design Variables

The values of any design variables in the circuit must be set before running a simulation. If values have been set in the past, they appear in the Value column of the Design Variables section of the Setup assistant in the ADE Explorer form.

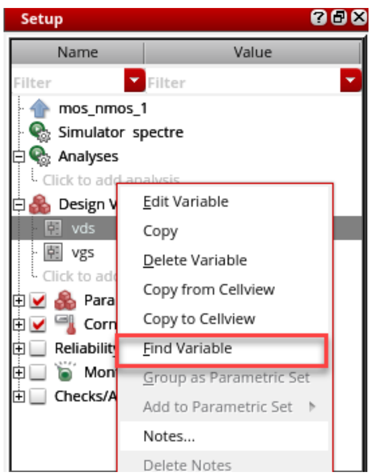
Note: If there are variables in the text files down in the hierarchy, retrieve them. Choose **Variables – Copy from Cellview** from the ADE Explorer menu.



Note: The *vgs*, *vds* variables appear in a table in the Design Variables section.



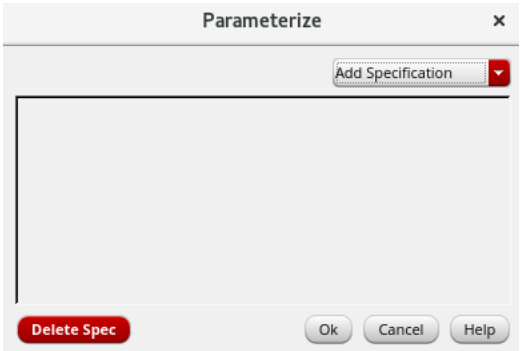
1. To locate the *vds* variable in the *nmos* design, **right**-click the *vds* name and select **Find Variable**



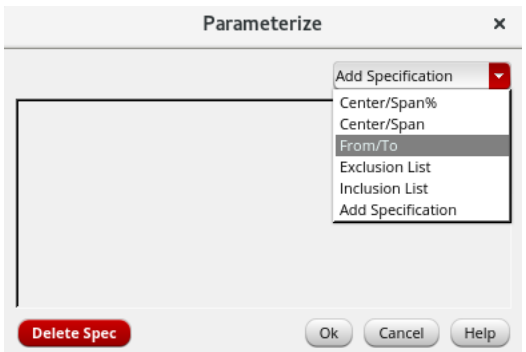
The Find Variable operation takes you to the *nmos* schematic tab and highlights the *vds* instance. Repeat the same by **right-**clicking on the **Find Variable** of *vgs* to highlight the *vgs*.

2. Move to the **maestro** tab and set the value of the *vds* variable to **5** by clicking the *Value* field after the variable *vds* and entering the new value. Press **Enter** to accept the change.

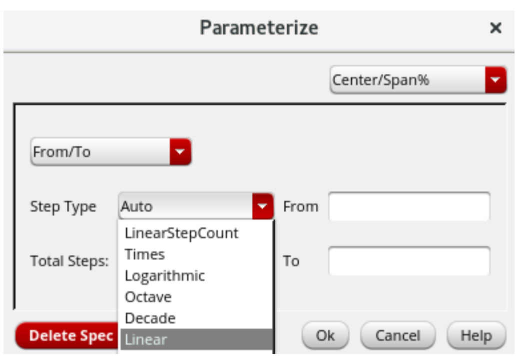
3. For the *vgs*, click on the *value* field to enable the button ... of the **Parameterize** form. Click on it to open the form once it is enabled.



4. In the drop-down of **Add Specification**, click on **From/To** to add the *Spec*.



5. In the **Step Type** drop-down menu, click on **Linear.**



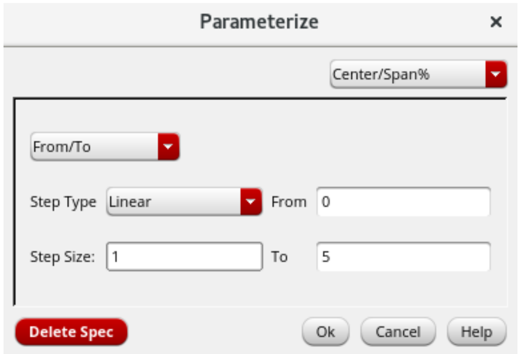
6. Type the values as follows:

   From: 0

   To: 5

   Step Size: 1

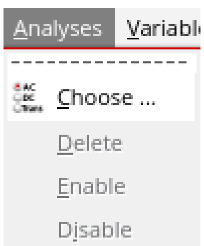
Your form should look similar to the one you see in the following snapshot. Click **OK.**



Choosing Analyses

This part demonstrates how to select **dc analyses** to complete your circuit during the simulation. You can select and run dc analysis on the *nmos* design.

1. In the Simulation window, choose **Analyses – Choose**.



2. Or, click the **Choose Analysis** icon. 

The Choosing Analyses form appears.

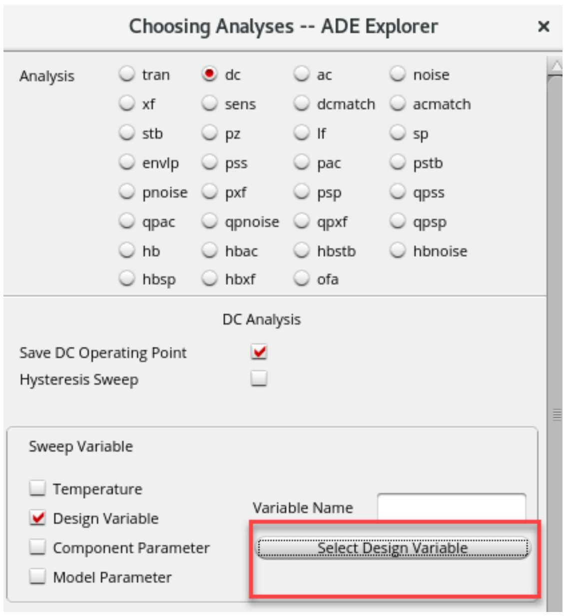
Take a few moments to become familiar with this form because it is used often throughout the lab activities. It is also a dynamic form; the layout changes based on the selected analyses.

3. Set up a DC analysis.

   a. In the Analysis section, select **dc**.

   b. In the DC analysis section, select the **Save DC Operating Point** checkbox to turn it on.

   c. In the **Sweep Variable** section, click on **Design Variable** to select the variable to be in the input or X-axis.

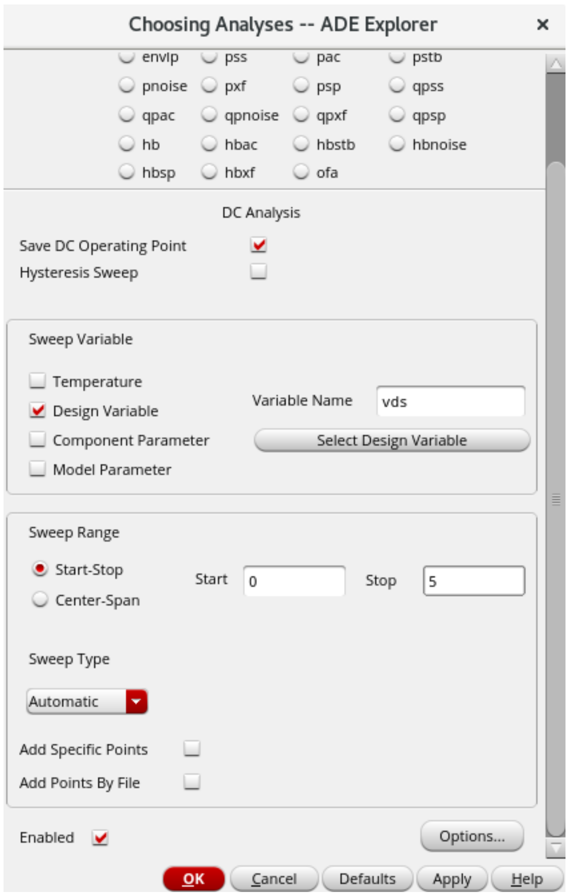


   d. Click on **Select Design Variable** button, and in the form opened, click on *vds* variable and click **OK.**

e. In the **Sweep Range** section, click on the *Start-Stop* radio button and give the values as follows:

Start: `0`, Stop: `5`

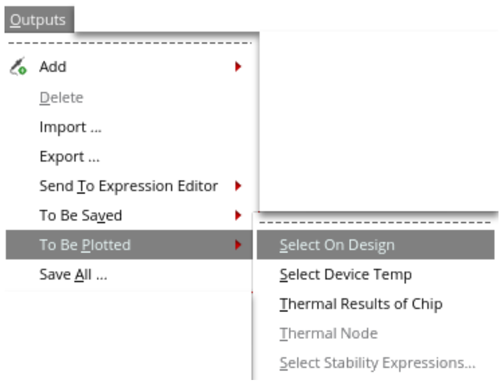
Ensure that the **Enabled** checkbox is selected. Your form should look similar to the one you see in the following snapshot. Click on **OK.**



Saving and Plotting Simulation Outputs

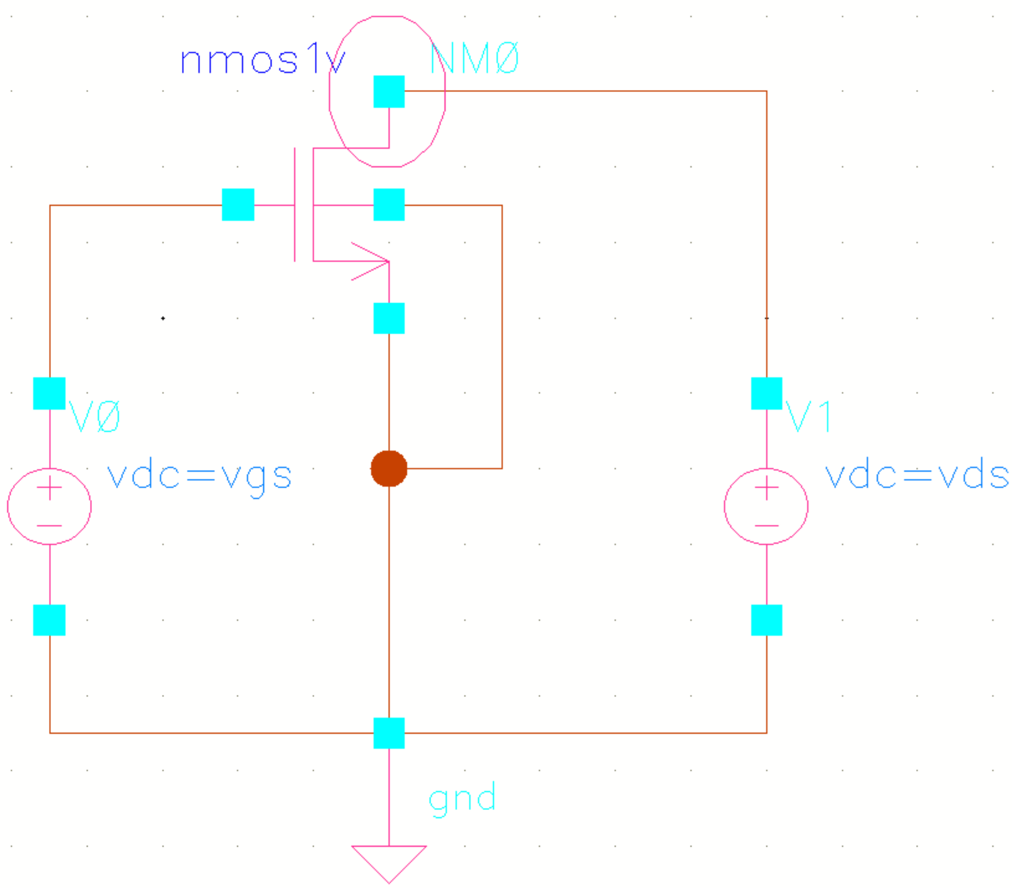
The simulation environment is configured to save all public node voltages by default. It also saves the currents you select for plotting and saving.

1. To select specific signals for saving and plotting, choose **Outputs – To Be Plotted – Select On Design** in the ADE Explorer window.



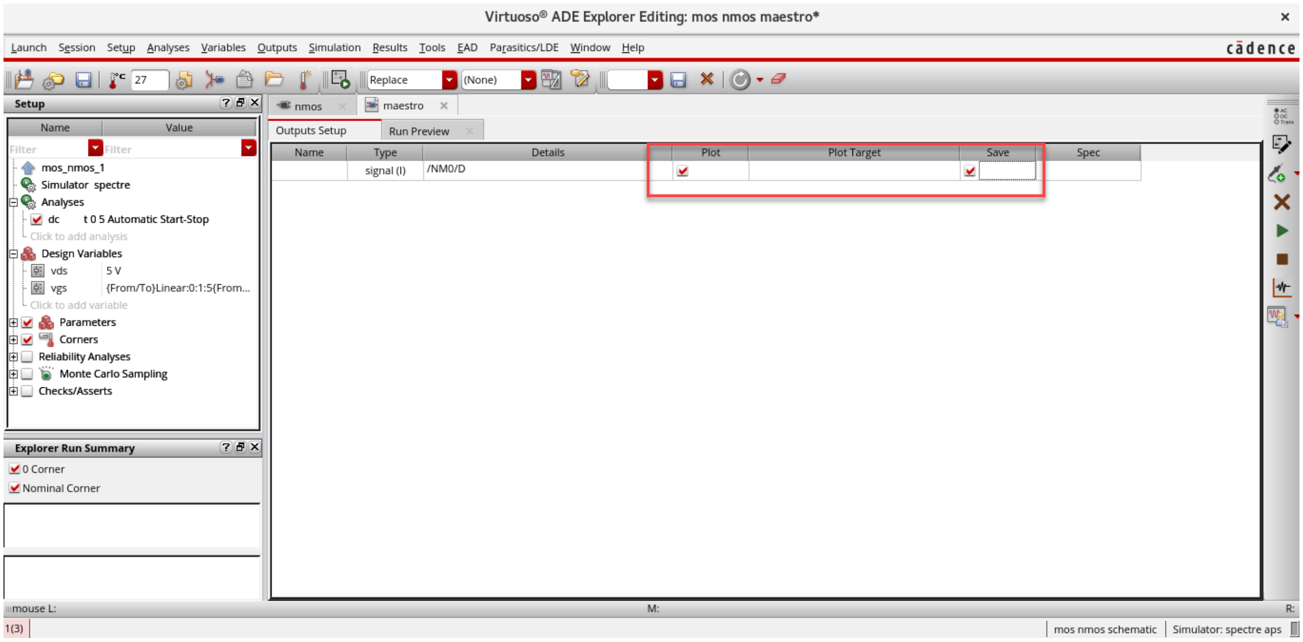
This brings you to the front of the schematic window for your selections.

2. Select the **drain** terminal for plotting. After selections, the terminal gets highlighted.



3. Press **Esc** with your cursor in the schematic window to stop the selection process.

4. Move to the **maestro** tab. Enable the plot and verify whether it looks like the one in the following snapshot:



Running the Simulation

1. Choose **Simulation – Netlist and Run or** click the **Run Simulation** button  to recreate the netlist and run the simulation.

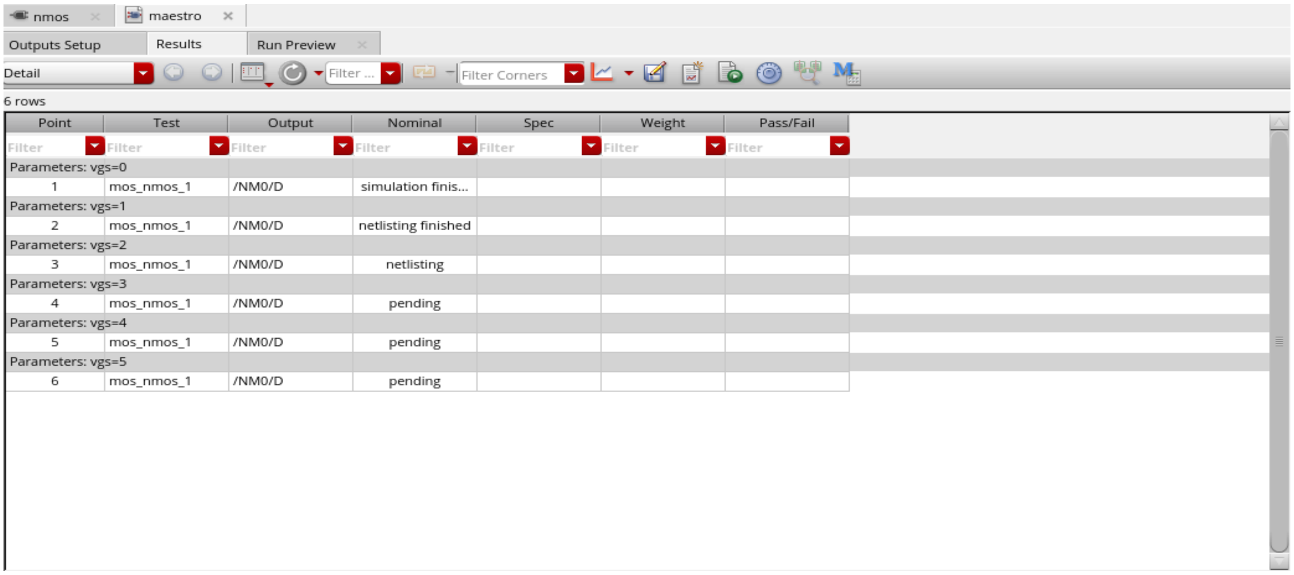
   a. As an alternative to the previous step, you can choose **Simulation – Run** from the menu if you have already created your netlist.

   b. A log file shows the progress of the simulation for parametric analyses of *vgs*. This is called the *spectre.out* file and is stored in a hierarchical psf directory inside your *./simulation* directory.

   c. The log file also lists warnings and errors if there are problems in the input data, which often is missing model files.

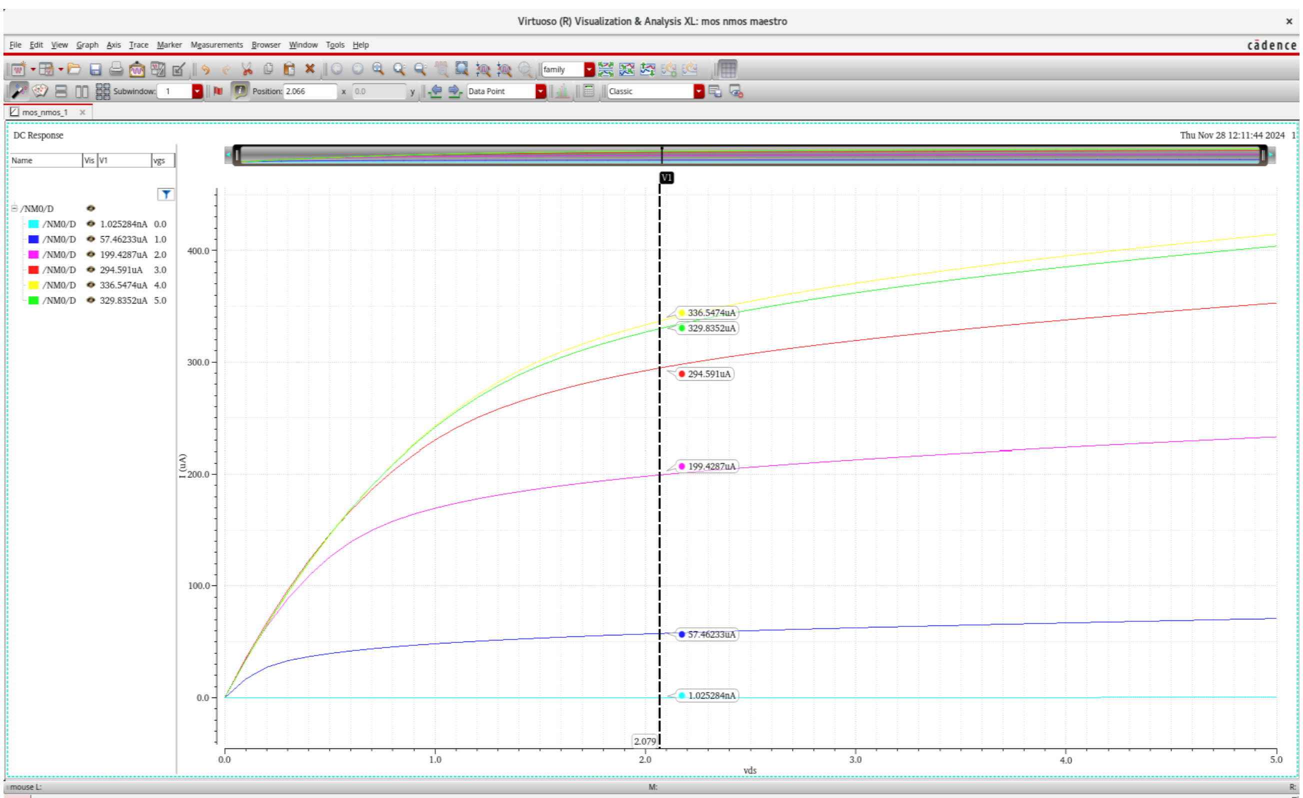
   Tip: If you close this log file, you can open it again by going to **Simulation – Output Log**.

2. In the Results tab of the Maestro, shows the status of each trace or value of *vgs*, according to the *Step Size* given in the Parameterize form.

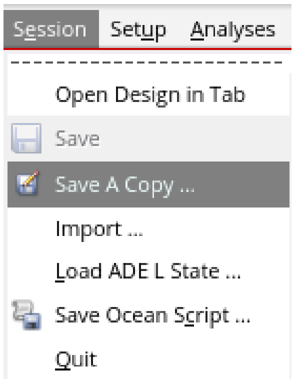


Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
Parameters: vgs=0						
1	mos_nmos_1	/NM0/D	simulation finis...			
Parameters: vgs=1						
2	mos_nmos_1	/NM0/D	netlisting finished			
Parameters: vgs=2						
3	mos_nmos_1	/NM0/D	netlisting			
Parameters: vgs=3						
4	mos_nmos_1	/NM0/D	pending			
Parameters: vgs=4						
5	mos_nmos_1	/NM0/D	pending			
Parameters: vgs=5						
6	mos_nmos_1	/NM0/D	pending			

3. When the simulation finishes, the DC Response plot is plotted automatically in the Virtuoso® Visualization & Analysis XL window.



4. Click on the **Save** icon to save the graph. Move to Maestro window and save the simulation settings by clicking on the **Session – Save a Copy.**



5. In the **Save a Copy** form, name the view and save it.

End of Lab

Module 2: Design and Characterization of a p-MOS Circuit

Lab 2-1 Design of a p-MOS Circuit

Objective: To use the basic schematic capture commands to build a p-MOS circuit.

In this lab, you open the schematic capture environment window and add components, wires and pins. The schematic is then checked and saved.

Creating a Library and Cellview

1. Open the Terminal window, open the directory of **basic_analog_labs,** by entering this command in the terminal window:

   ```
   cd basic_analog_labs
   ```

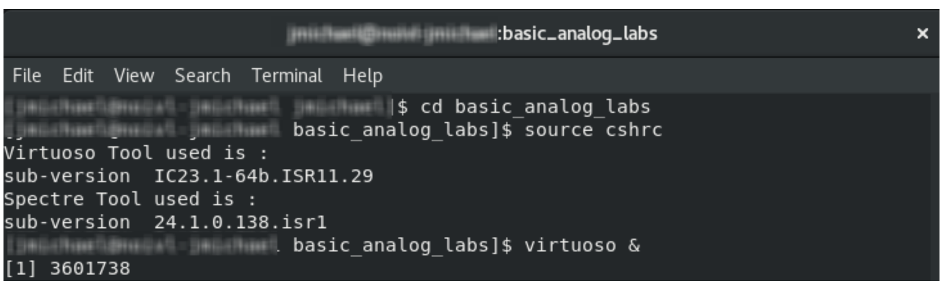
2. In that directory, source the License file, by entering this command in the terminal window:

   ```
   source cshrc
   ```

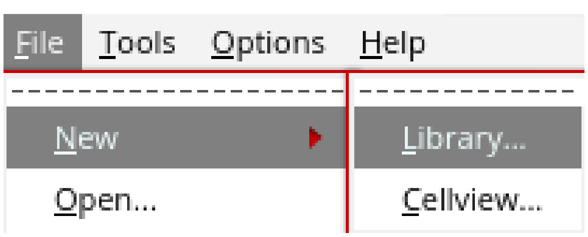
3. Type the following command to open CIW:

   ```
   virtuoso &
   ```

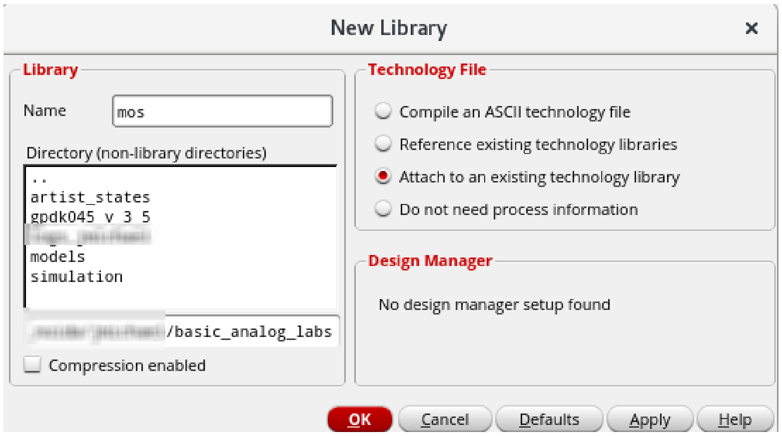
   The following screenshot gives the view of the terminal window, on these commands executed:

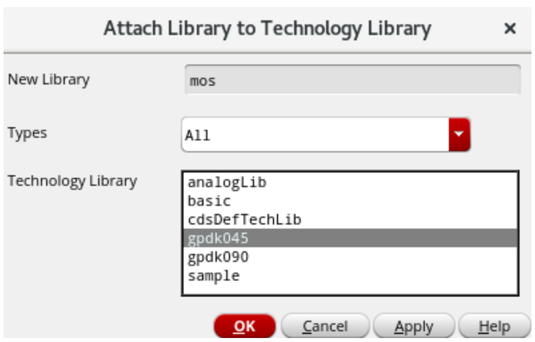
4. Open CIW, and from CIW, choose **File – New – Library**.

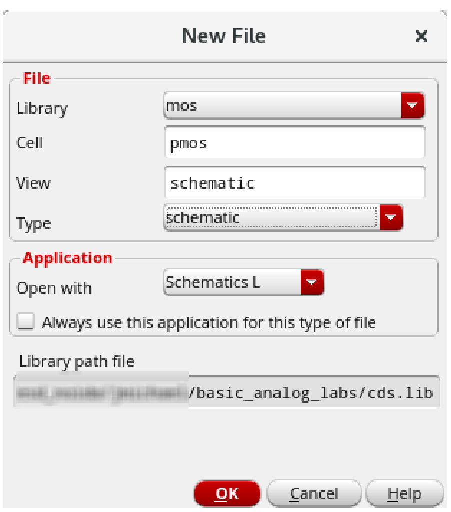
5. Name the Library as **mos** and select the radio button for **Attach to an existing technology Library** and click **OK.**



6. In the Attach Library to Technology Library form, select **gpdk045** library, and click **OK**.



7. Create a new cell view under the same **mos** library created, name it **pmos**, and click **OK** to open the schematic view.



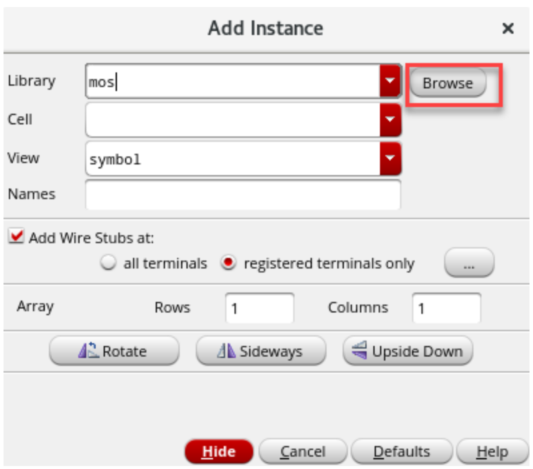
Invoking the Add Instance Form in the Cellview

The instructions in this section describe how to build the p-mos circuit schematic shown in the following snapshot.

1. With your cursor in the schematic window, go to the top of the schematic window to the toolbar. Click the **Create Instance** icon to display the Add Instance form. 

Note: You can also choose **Create – Instance** from the pull-down menus or use the **I** key to pop up the form.

2. In the **Add Instance** form, click on **Browse**, to open the **Library Browser**.



3. In the **Library Browser – Add Instance** form, select **gpdk045** in the library column.

   The Cell column of this form now displays the available cells in that library.

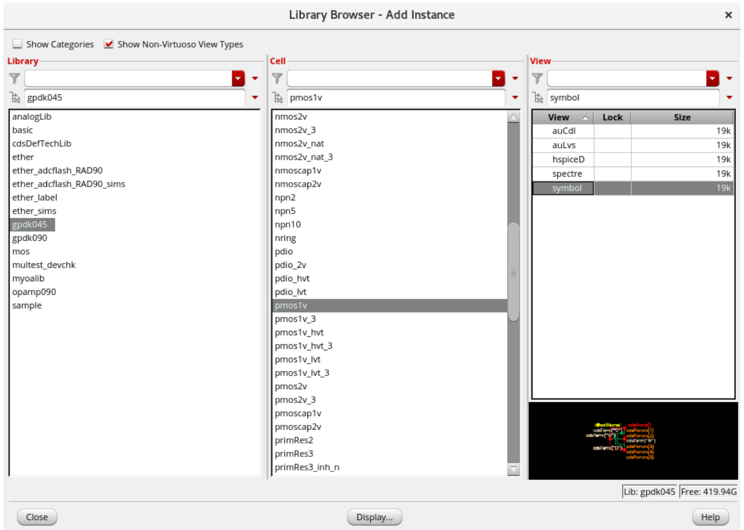
   a. Use the scroll bar to the right of the Cell column to locate **pmos1v** and **click** to select this cell. The Library Browser form now displays several views for the p*mos1v* cell.

      Note: You can use the filtering option under the Cell category.

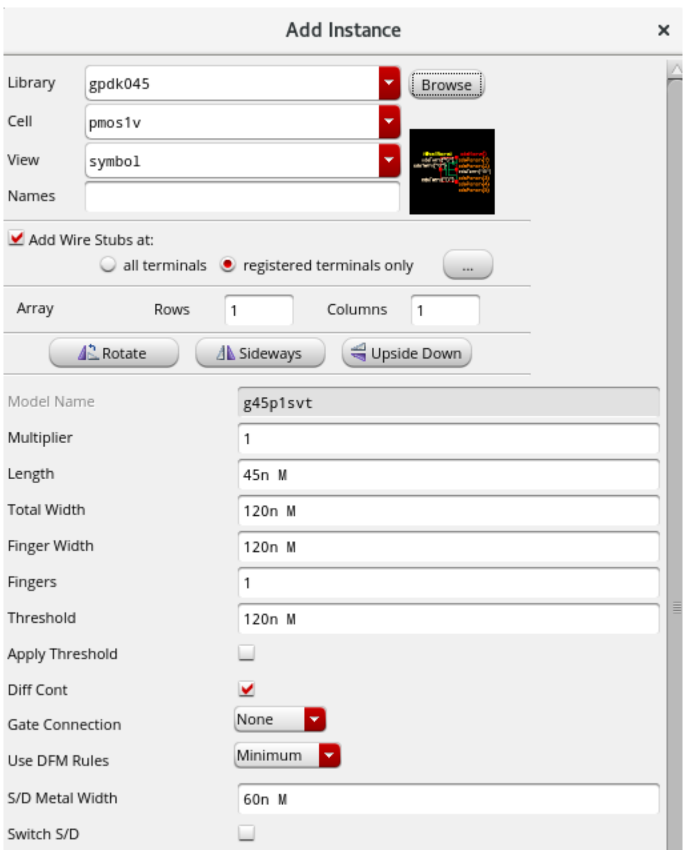
      Note: You can also begin typing ***pmos1v*** in the *cell* name field of the Add Instance form. The auto-complete feature brings the *pmos1v* cell name into the field for easy selection.

      As you make your selection, note that the Add Instance form modifies itself.

b. Select the **symbol view** with the left mouse button. Your Library Manager should look like the following snapshot.



The Add Instance form now displays some information along with auto-filled text fields, as shown in the following snapshot.



4. Close the **Library Browser – Add Instance** form. Leave the Add Instance form open with *Library*, *Cell* and *View* pull-down cyclic fields populated.

Adding Components to a Schematic

1. Click the **Hide** button toward the bottom of the form.

   Follow the next steps diligently, as they will allow you to place the chosen *pmos* transistor on your schematic.

   a. Move your cursor into the schematic window. A highlighted symbol is displayed. It is ready to be placed on the schematic.

   b. **Left**-click to place the symbol on the schematic.

   c. Place the instance of *pmos1v* on the schematic.

      Note: Observe that the device has been added to the schematic and the symbol has an instance name. A highlighted symbol of *nmos1v* remains and moves with the cursor.

   d. Click the <**Esc**> key to quit placement mode.

2. Press the **I** key.

   The Add Instance form is displayed again, with the library, cell, and view fields filled. Likewise, a highlighted symbol tracks the cursor in the schematic window.

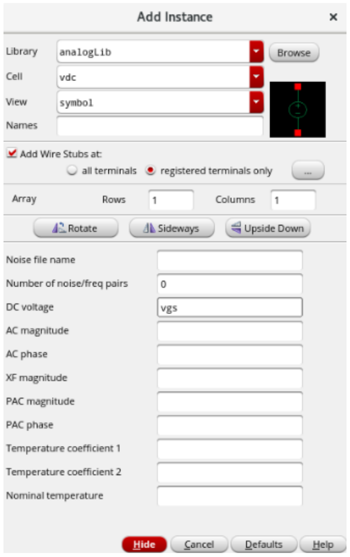
   a. Move the cursor into the Add Instance form. Change the library field to **analogLib** from the library column. Click in the cell text field and choose **vdc**.

      Note: You can use the **Browse** button to search for *vdc*. However, entering *vdc* directly into the text field saves time.

   b. Set the parameter as follows:

      DC voltage: `vgs`

c. Your form should be similar to what you see in the following snapshot:



d. **Hide** the form. Place this instance of *vdc* on the schematic, on left side of *pmos1v* instance.

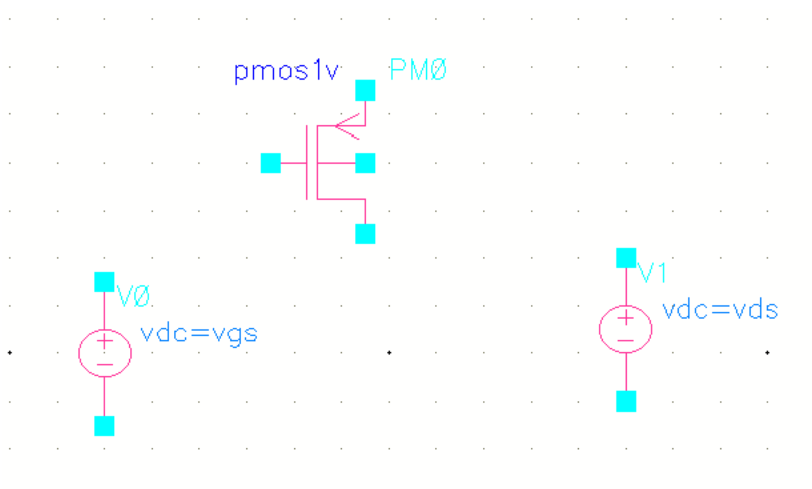
3. Press the **I** key.

   With the already selected **vdc** in the Add Instance form, change the parameter as follows:

   DC voltage: `vds`

   Hide the form. Place this instance of *vdc* on the schematic on the right side of the *pmos1v* instance.

4. You can use the keyboard arrow keys to move up and down the schematic. Your canvas should be similar to what you see in the following snapshot.



Adding Ground to the Schematic

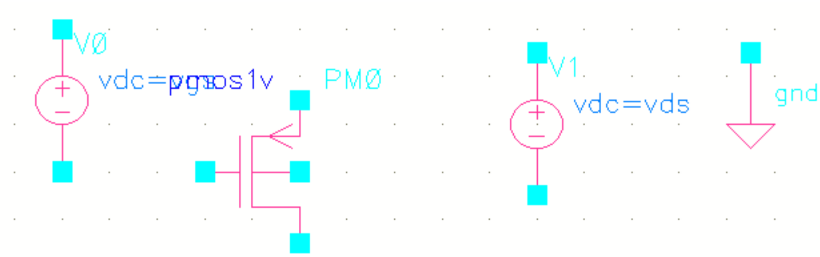
1. Zoom out with the **[** key, so you have room to place the power and ground symbols.

2. Press **I** to display the Add Instance form.

3. In the ***Cell*** text field of the Add Instance form, replace *vdc* with **gnd** under the same *analogLib* library.

4. Move the cursor into the schematic and place the *gnd* symbol below the source of the *pmos1v* device. Then press the **Esc** key. This will *close* the Add Instance form.

Important: Pressing **Esc** cancels any active command in the schematic window.



Adding Wires to a Schematic

Add wires to connect components in the design.

1. Choose **Create – Wire (narrow)** from the pull-down menu. You can also use the **Create Narrow Wire** button from the Create toolbar section.

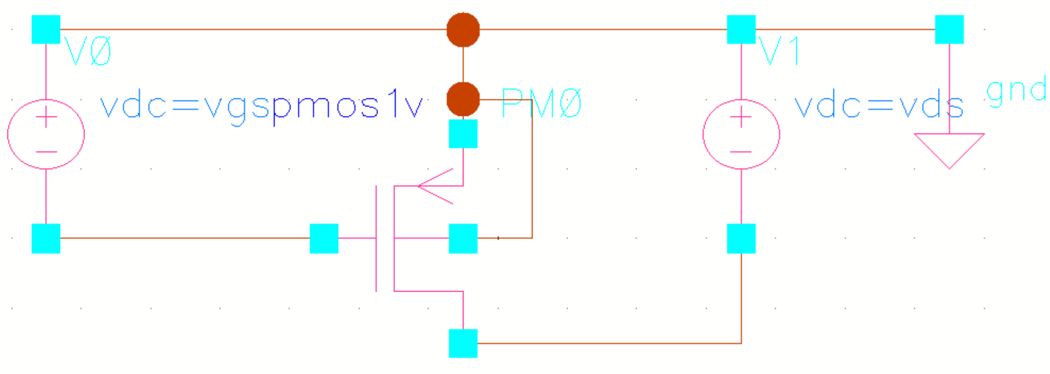
   Note: Lowercase **w** is the bindkey for a *Narrow wire*, **Shift-W** for a *Wide wire*.

2. In the schematic window, click one of your components' terminals, such as the *vdc* negative terminal, to make it the first point for your wiring.

   Note: The wire router cannot always find a path between two points. If this happens, a dotted flight line is drawn between the points and establishes connectivity. If flight lines occur in your lab, delete them and click on the intermediate points to guide the router as your design is wired.

   Follow the prompts at the bottom of the design window, move the mouse cursor and **left-**click the destination point for your wire, such as pmos gate. A net is routed.

3. Continue wiring the schematic, as shown in the following snapshot.



   The exact topology shown in the reference schematic is not as important as the connectivity.

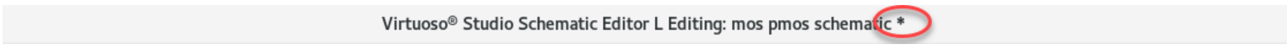
   Caution: *Be sure to connect the substrate for the pmos1v transistor to gnd respectively.*

   Important: If you make a mistake while wiring, select that section of the wire and press the **<Delete>** key to delete that part of the wire.

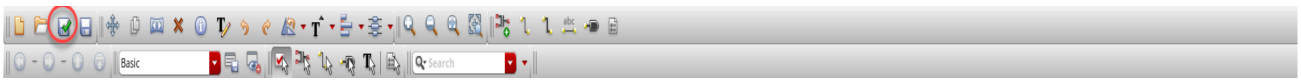
4. Press **Esc** with your cursor in the schematic window to cancel the wiring when finished wiring.

Saving a Design

Note on the title bar that there is an asterisk next to the name schematic. This indicates that your schematic has been edited and needs to be saved.

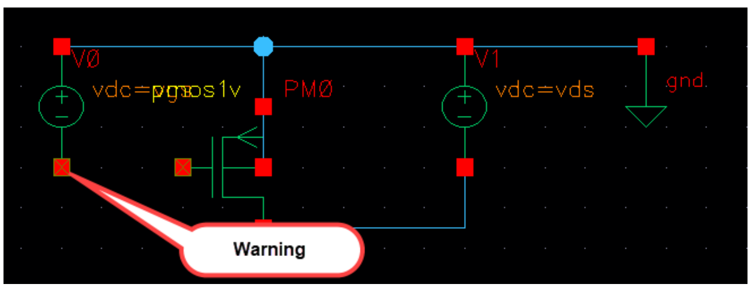


1. Click the **Check and Save** icon in the schematic editor window.



2. Or, choose **File – Check and Save** or click **Shift + X**.

Note: A yellow box appears around all the unconnected pins and wires. If there are any, close the associated warning message and complete the connectivity.

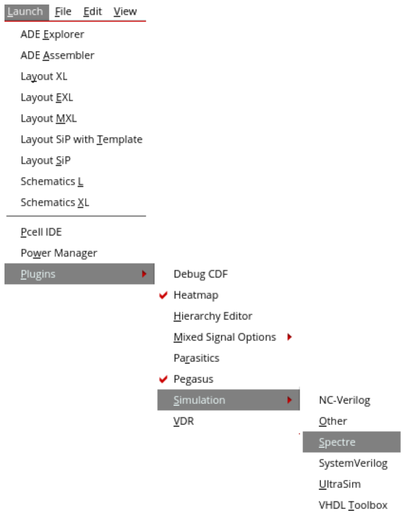


3. Check the CIW output area for any messages after a successful check and save.

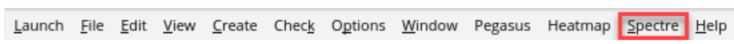
```
INFO (SCH-1170): Extracting "pmos schematic"
INFO (SCH-1426): Schematic check completed with no errors.
INFO (SCH-1181): "mos pmos schematic" saved.
```


Netlisting the Design

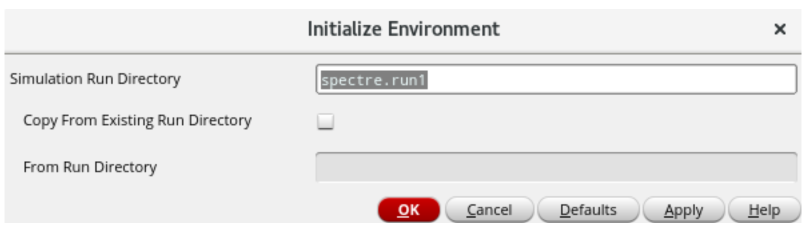
1. From the Schematic Editor window, choose **Launch – Plugins – Simulation – Spectre**.



Note: This attaches the Spectre Classic Simulator in the menu bar.

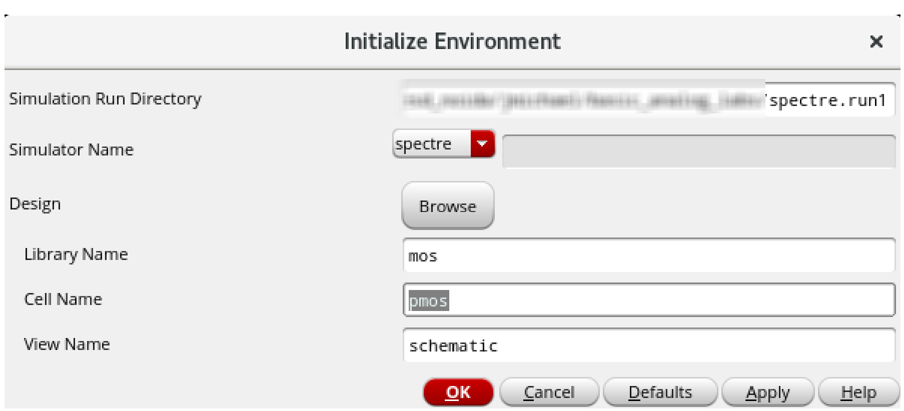


2. From the menu bar, choose **Spectre – Initialize**.

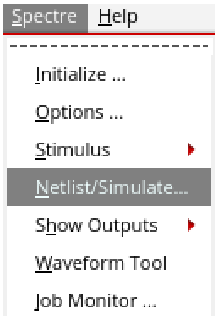


3. Click **OK** in the Initialize Environment form to create simulation result databases in the *spectre.run1* default simulation run directory.

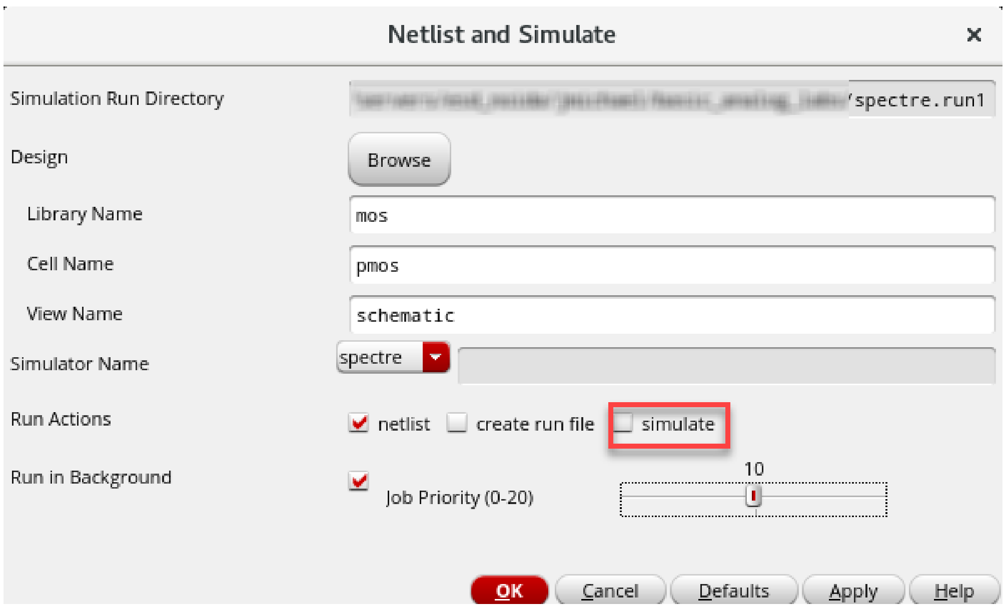
   a. Click **OK** in the next form that opens (shown in the following snapshot), verifying the design to be netlisted.



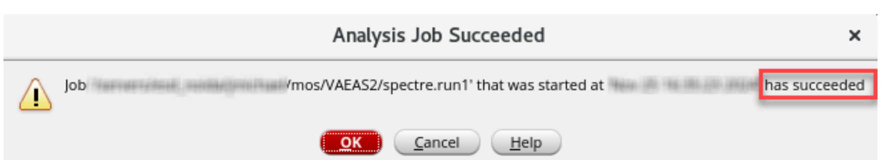
4. Choose **Spectre – Netlist/Simulate** to open the Netlist and Simulate form.



5. **Disable** the *Run Actions – simulate* option. Your form should look similar to the one you see in the following snapshot.



6. Click **OK**. In the *Analysis Job Succeeded* form that comes up, click **OK**.



7. View the netlist file in the *spectre.run1* directory by choosing **Spectre – Stimulus – Edit Netlist File**.

```
// Library name: mos
// Cell name: pmos
// View name: schematic
PM0 (net4 net2 0 0) g45p1svt w=(120n) l=45n nf=1 as=16.8f ad=16.8f ps=520n \
        pd=520n nrd=1.16667 nrs=1.16667 sa=140n sb=140n sd=160n \
        sca=226.00151 scb=0.11734 scc=0.02767 m=(1)
V1 (0 net4) vsource dc=vds type=dc
V0 (0 net2) vsource dc=vgs type=dc
```

Try to associate the connectivity in the netlist along with your drawn schematic.

8. **Close** the netlist window, leave *nmos* schematic view and CIW session running and proceed to the next lab.

End of Lab

Lab 2-2 Simulation Using ADE Explorer and Virtuoso VA

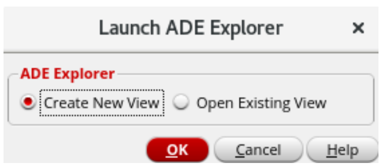
Objective: To simulate the p-MOS circuit using ADE Explorer and Virtuoso VA.

In this lab, you

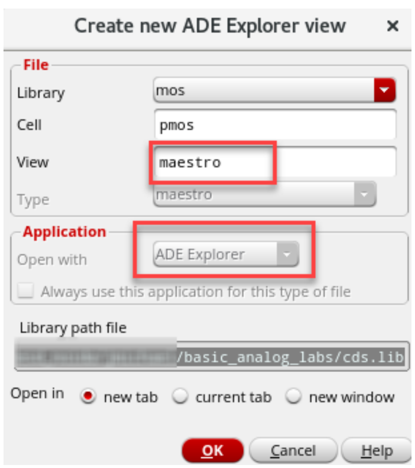
- Learn how to simulate the p-MOS created in the previous lab by creating a new *maestro view* using Virtuoso® ADE Explorer.

Starting the Virtuoso ADE Explorer and Creating a *maestro* View

1. Choose **Launch – ADE Explorer** from the *pmos* schematic window to open the Launch ADE Explorer form.

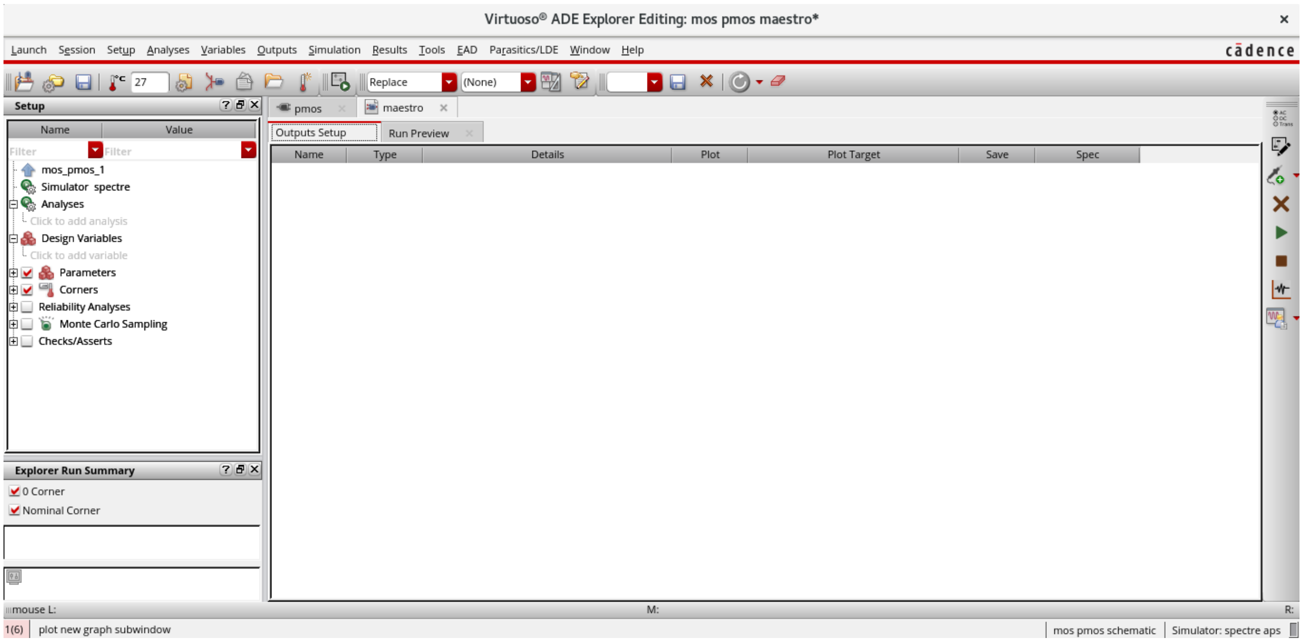


2. Select **Create New View** to open the **Create new ADE Explorer view** form.



3. The **maestro** view type is chosen by default, and so is the *Open with* the application, which is set to the **ADE Explorer**.

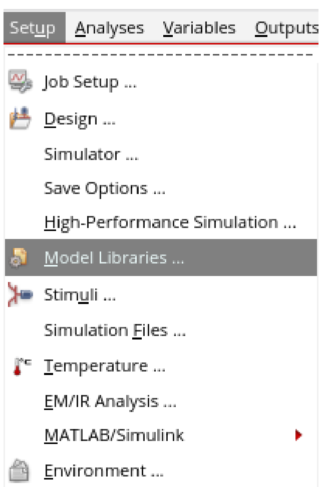
   Leave the View name set to the default name, **maestro**. Choose to open in a new tab and click **OK** on this form to open the ADE Explorer environment in a new **maestro** tab.



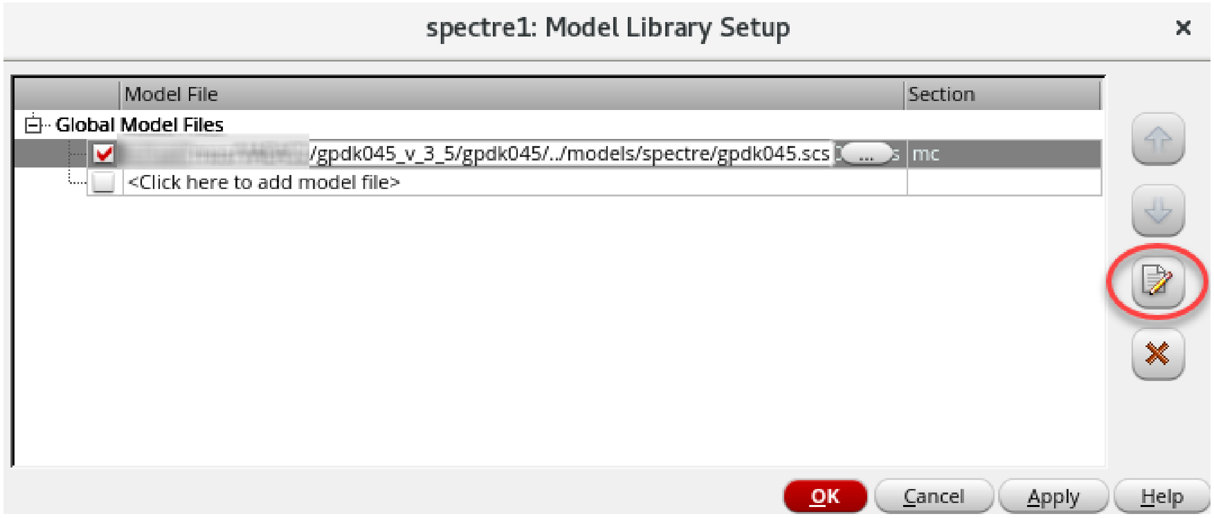
Setting the Model Libraries

The model library file contains the model files that describe the *pmos1v* device during simulation.

1. Choose **Setup – Model Libraries**.



The Model Library Setup form appears. You can see that there will be a *gpdk045 model library* added automatically. This is because we have attached the **mos** library to *gpdk045* in the beginning.



a. To view the model file, select the model file. Click the **Edit/View Selected Files** button.

Close the model file after viewing.

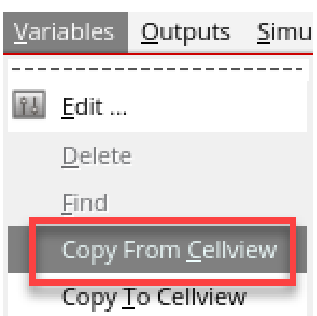
With the Model Library Setup form, you can include multiple models.

b. On the Model Library Setup form, click **OK.**

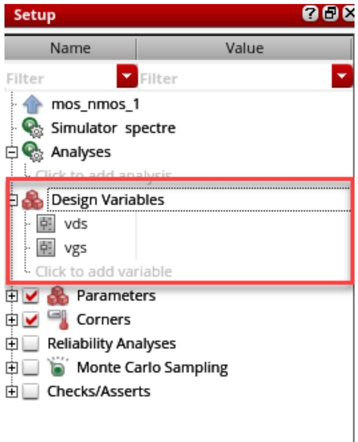
Setting Design Variables

The values of any design variables in the circuit must be set before running a simulation. If values have been set in the past, they appear in the Value column of the Design Variables section of the Setup assistant in the ADE Explorer form.

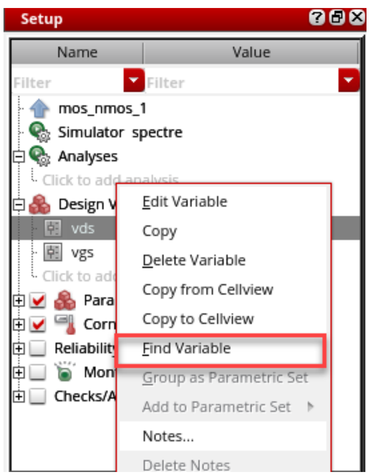
Note: If there are variables in the text files down in the hierarchy, retrieve them. Choose **Variables – Copy from Cellview** on the ADE Explorer menu.



Note: The *vgs*, *vds* variables appears in a table in the Design Variables section.



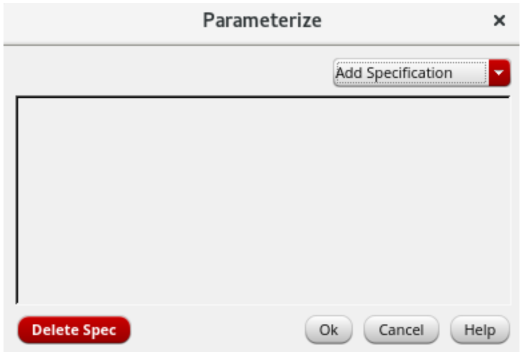
1. To locate the *vds* variable in the *pmos* design, **right**-click the *vds* name and select **Find Variable**.



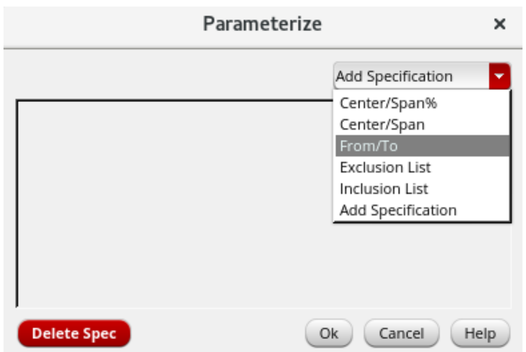
The Find Variable operation takes you to the *pmos* schematic tab and highlights the *vds* instance. Repeat the same by **right-**clicking on the **Find Variable** of *vgs* to highlight the *vgs*.

2. Move to the **maestro** tab and set the value of the *vds* variable to **5** by clicking the *Value* field after the variable *vds* and entering the new value. Press **Enter** to accept the change.

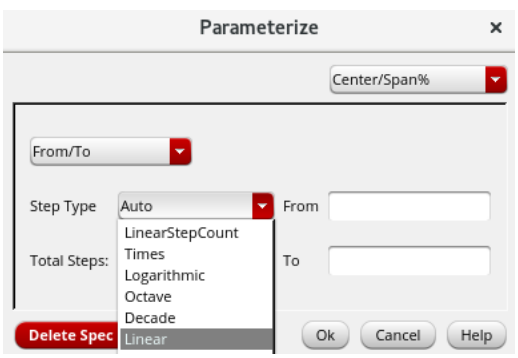
3. For the *vgs*, click on the *value* field to enable the button  on the **Parameterize** form. Click on it to open the form once it is enabled.



4. In the **Add Specification** drop-down menu, click on **From/To** to add the *Spec.*



5. In the **Step Type** drop-down menu, click on **Linear.**



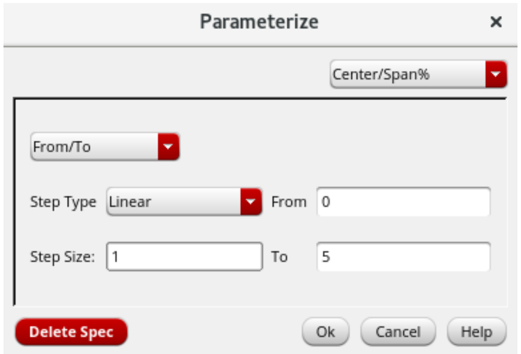
6. Type the values as follows:

   From: 0

   To: 5

   Step Size: 1

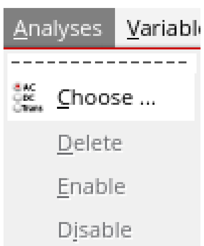
Your form should look similar to the one you see in the following snapshot. Click **OK.**



Choosing Analyses

This part demonstrates how to select **dc analyses** to complete your circuit during the simulation. You can select and run DC analysis on the *pmos* design.

1. In the Simulation window, choose **Analyses – Choose**



2. Or, click the **Choose Analysis** icon. 

The Choosing Analyses form appears.

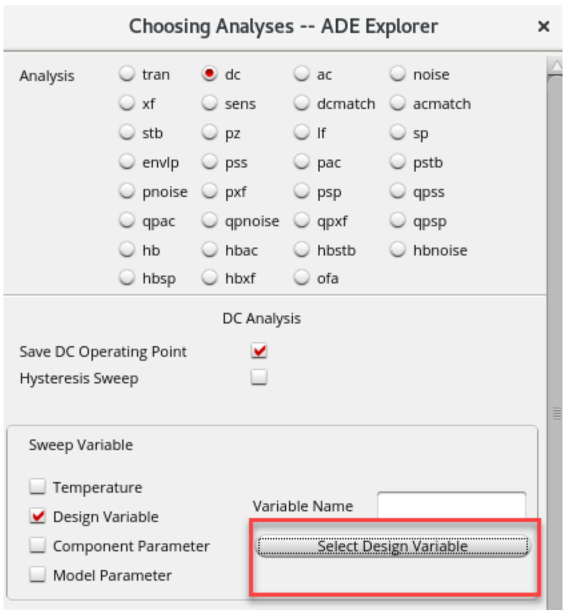
Take a few moments to become familiar with this form because it is used often throughout the lab activities. It is also a dynamic form; the layout changes based on the selected analyses.

3. Set up a DC analysis.

   a. In the Analysis section, select **dc**.

   b. In the DC analysis section, select the **Save DC Operating Point** checkbox to turn it on.

   c. In the **Sweep Variable** section, click on **Design Variable** to select the variable to be in the input or X-axis.

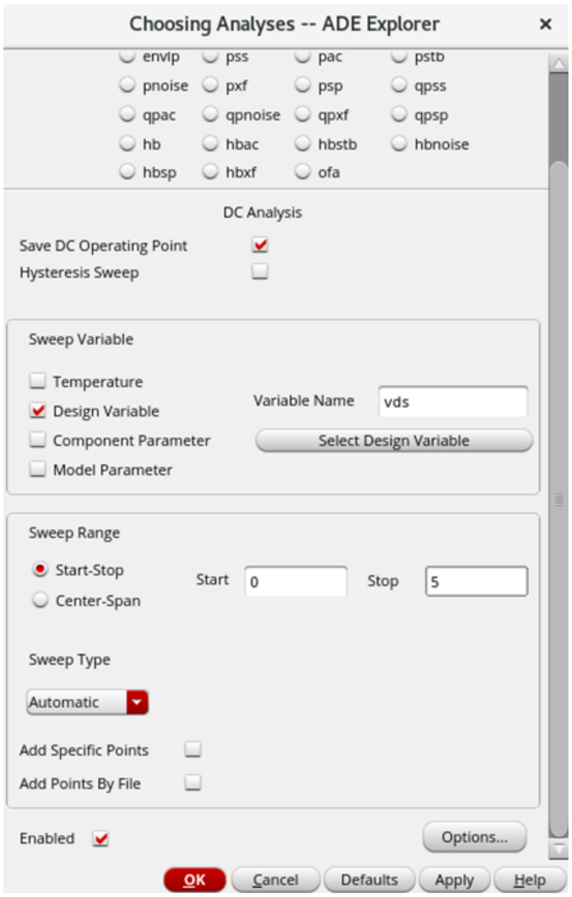


   d. Click on the **Select Design Variable** button, and in the form opened, click on the *vds* variable and click **OK.**

e. In the **Sweep Range** section, click on the *Start-Stop* radio button and give the values as follows:

Start: `0`, Stop: `5`

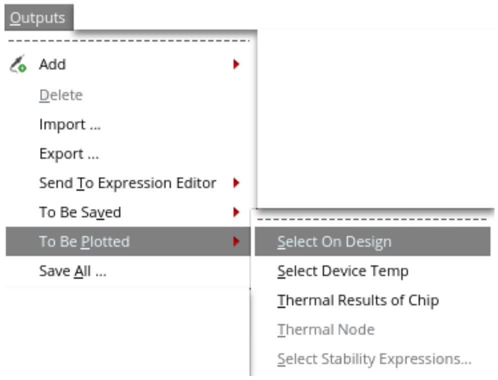
f. Ensure that the **Enabled** checkbox is selected. Your form should look similar to the one you see in the following snapshot. Click **OK.**



Saving and Plotting Simulation Outputs

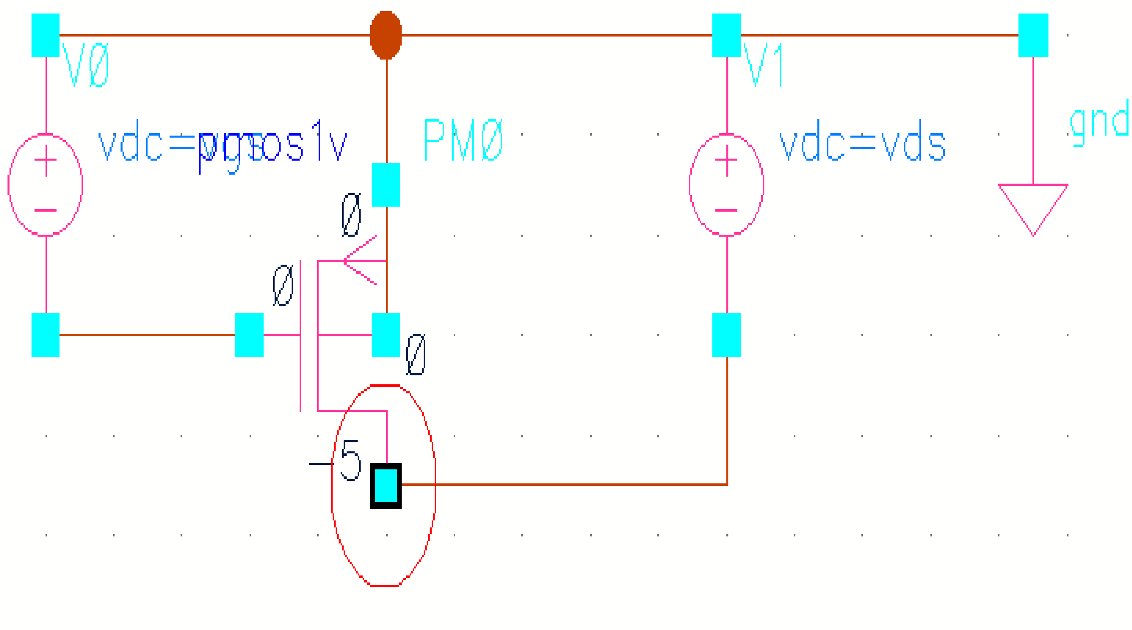
The simulation environment is configured to save all public node voltages by default. It also saves the currents you select for plotting and saving.

1. To select specific signals for saving and plotting, choose **Outputs – To Be Plotted – Select On Design** in the ADE Explorer window.



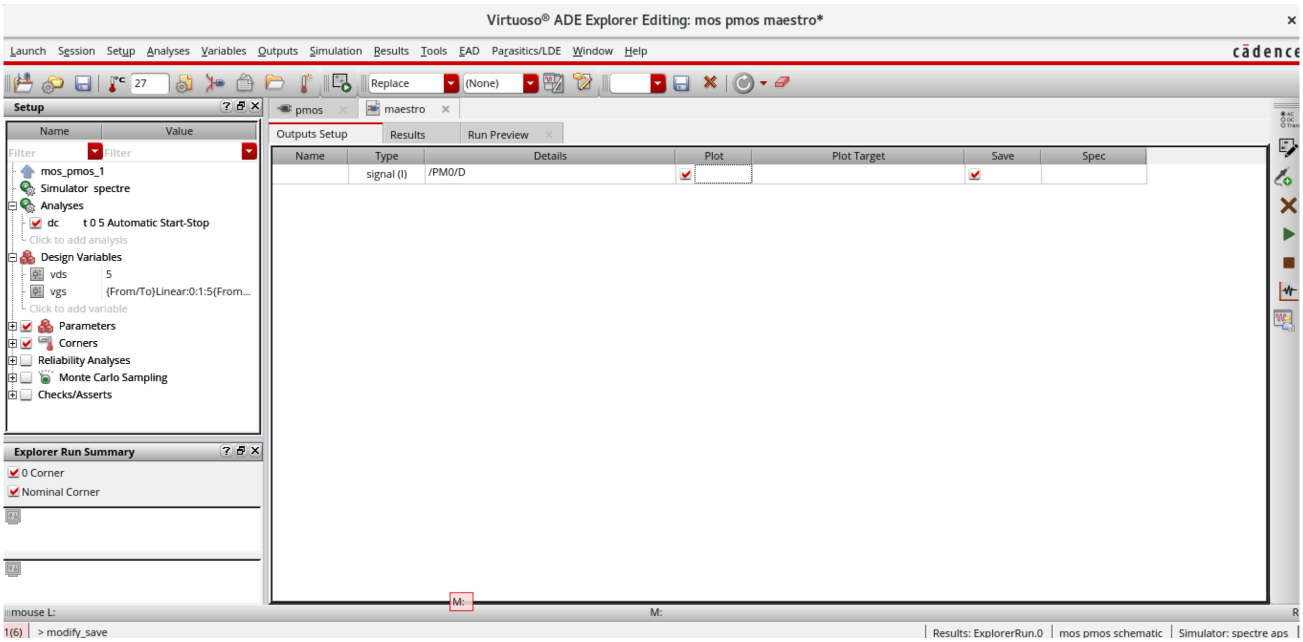
This brings you to the front of the schematic window for your selections.

a. Select the *drain* terminal for plotting. After selections, the terminal gets highlighted.



2. Press **Esc** with your cursor in the schematic window to stop the selection process.

3. Move to the **maestro** tab. Enable the plot and verify whether it looks like the one in the following snapshot:



Running the Simulation

1. Choose **Simulation – Netlist and Run or** click the **Run Simulation** button  to recreate the netlist and run the simulation.

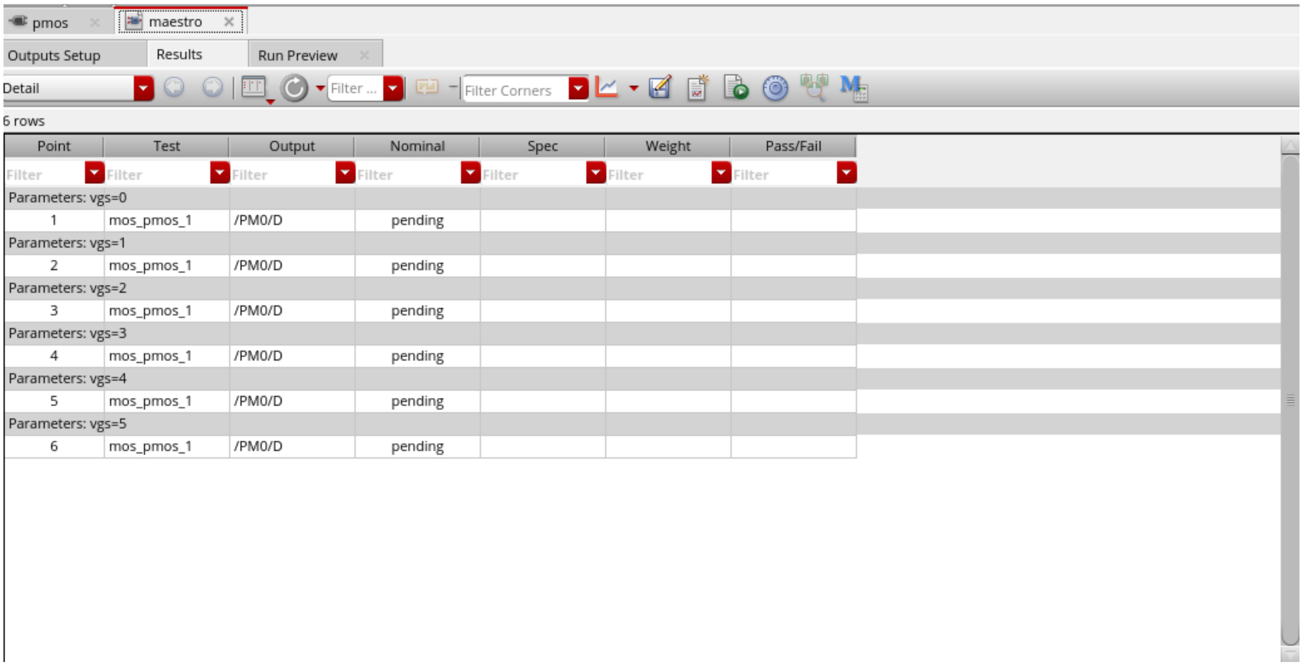
   a. As an alternative to the previous step, you can choose **Simulation – Run** from the menu if you have already created your netlist.

   b. A log file shows the progress of the simulation for parametric analyses of *vgs*. This is called the *spectre.out* file and is stored in a hierarchical psf directory inside your *./simulation* directory.

   c. The log file also lists warnings and errors if there are problems in the input data, which often is missing model files.

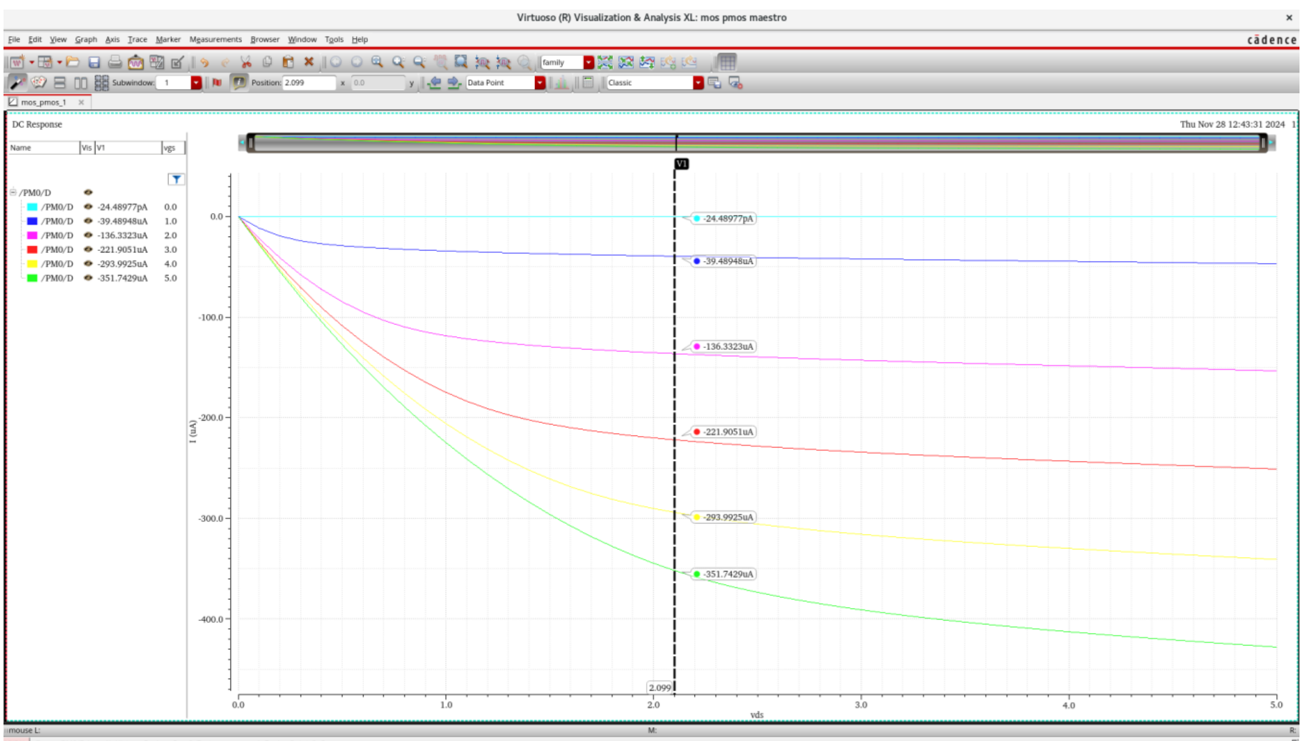
   Tip: If you close this log file, you can open it again by going to **Simulation – Output Log**.

2. In the Results tab of the Maestro, shows the status of each trace or value of *vgs*, according to the *Step Size* given in the Parameterize form.

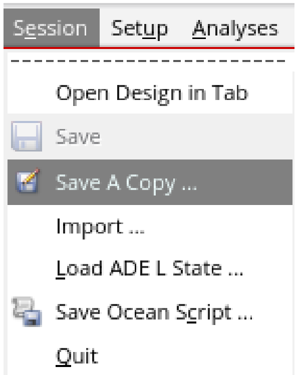


Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
Parameters: vgs=0						
1	mos_pmos_1	/PM0/D	pending			
Parameters: vgs=1						
2	mos_pmos_1	/PM0/D	pending			
Parameters: vgs=2						
3	mos_pmos_1	/PM0/D	pending			
Parameters: vgs=3						
4	mos_pmos_1	/PM0/D	pending			
Parameters: vgs=4						
5	mos_pmos_1	/PM0/D	pending			
Parameters: vgs=5						
6	mos_pmos_1	/PM0/D	pending			

3. When the simulation finishes, the DC Response plot is plotted automatically in the Virtuoso® Visualization & Analysis XL window.



4. Click on the **Save** icon to save the graph. Move to Maestro window and save the simulation settings by clicking on the **Session – Save a Copy.**



5. In the **Save a Copy** form, name the view and save it.

End of Lab